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MTDE: An effective multi-trial vector-based differential evolution algorithm and its applications for engineering design problems

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ABSTRACT

In this article, an effective metaheuristic algorithm named multi-trial vector-based differential evolution (MTDE) is proposed. The MTDE is distinguished by introducing an adaptive movement step designed based on a new multi-trial vector approach named MTV, which combines different search strategies in the form of trial vector producers (TVPs). In the developed MTV approach, the TVPs are applied on their dedicated subpopulation, which are distributed by a winner-based distribution policy, and share their experiences efficiently by using a life-time archive. The MTV can be deployed by different types of TVPs, particularly, we use the MTV approach in the MTDE algorithm by three TVPs: representative based trial vector producer, local random based trial vector producer, and global best history based trial vector producer. Therefore, this study introduces the MTV approach to boost the performance of the MTDE and demonstrates its advantages in dealing with problems of different levels of complexity. The performance of the proposed MTDE algorithm is evaluated on CEC 2018 benchmark suite which include unimodal, multimodal, hybrid, and composition functions and four complex engineering design problems. The experimental and statistical results are compared with state-of-the-art metaheuristic algorithms: GWO, WOA, SSA, HHO, CoDE, EPSDE, QUATRE, and MKE. The results demonstrate that the MTDE algorithm shows improved performance and benefits from high accuracy of optimal solutions obtained.

1. Introduction

Metaheuristics are successful alternatives for exact methods to solve the real-life optimization problems in polynomial time [1,2]. In the literature, a wide variety of metaheuristics have been introduced for continuous optimization such as differential evolution (DE) [3], particle swarm optimization (PSO) [4], krill herd (KH) [5], grey wolf optimizer (GWO) [6] and harris hawks optimization (HHO) [7]. In addition, different methods have been employed to develop the binary version of these continuous algorithms [8] to solve the real-life problems that have binary search space. Due to superior performance of metaheuristics, they have been widely used to solve the challenging problems in diverse fields such as disease diagnosis [9–11], task scheduling [12] and virtual machine placement [13] in cloud computing, chip design [14], tour planning [15], feature selection [16–21], engineering optimization [22–25] and financial issues such as predicting the digital currency price [26].

In the literature, the nature-inspired metaheuristic algorithms are mostly classified based on their inspiration into three categories: swarm intelligence, physics-based, and evolutionary algorithms [27–29]. To illustrate and increase the understanding these categories, a demographic view of their algorithms is shown in Fig. 1. The swarm intelligence algorithms (SIs) are inspired by the collective behavior of social creatures from insect colonies to human societies. They can find optimal solutions by cooperation, interaction, and competition. There have been proposed many swarm intelligence algorithms such as particle swarm optimization (PSO) [4], ant colony optimization (ACO) [30], whale optimization algorithm (WOA) [31], salp swarm algorithm (SSA) [32], emperor penguin optimizer (EPO) [33], conscious neighborhood-based crow search algorithm (CCSA) [34], marine predators algorithm (MPA) [35], and improved grey wolf optimizer (I-GWO) [36].

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This category may encounter some shortages like local optima trapping, premature convergence, and imbalance between the exploration and exploitation [37–39].

Physics-based algorithms mimic the physical rules in nature. In these algorithms, individuals communicate around the search space using concepts and laws of physics such as gravitational force, inertia force, light refraction law, and molecular dynamics. Some popular algorithms in this category are gravitational search algorithm (GSA) [40], henry gas solubility optimization (HGSO) [41], atom search optimization (ASO) [42], charged system search (CSS) [43], black hole (BH) [44], galaxy-based search algorithm (GBSA) [45], ray optimization (RO) [46], big bang-big crunch (BB-BC) [47], water cycle algorithm (WCA) [48], and equilibrium optimizer (EO) [49].

Evolutionary algorithms (EAs) are mostly inspired by the biological evolution process of species [50]. EAs are well known for their ability to handle nonlinear and complex optimization problems. The main advantage of EAs over other conventional optimization algorithms is that they do not require any additional information about the objective functions, such as differentiability or continuity [51]. The most popular algorithms in this category are genetic algorithm (GA) [52], which works based on the principle of the Darwin theory of survival, evolution strategy (ES) [53], evolutionary programming (EP) [54], genetic programming (GP) [55], and differential evolution (DE) [3]. Similarly to other EA algorithms, DE uses mutation and recombination operators to evolve the population, but it uses a difference vector and a parent–offspring competition to determine whether the offspring solution survives to the next generation or not [56,57]. The canonical DE has four controlling parameters, the number of generations, population size, scaling factor, and crossover rate, which remain unchanged during the optimization. Also, it has only one search strategy called mutation strategy, which uses three random individuals to produce a mutated vector.

In some variants of the DE algorithm, such as monkey king evolutionary algorithm (MKE) [58] and quasi-affine transformation evolution algorithm (QUATRE) [59], a constant value for control parameter scaling factor (F) is considered for all individuals. While they introduced a new matrix instead of random values of crossover rate (Cr). In addition, there have been proposed DE algorithms such as FADE [60], GADE [61], and EADE [62] by using different adaptation updating for some or all parameters during the evolution. Success-History based parameter adaptation for DE (SHADE) [63] is an improved version of the JADE [64] algorithm, which instead of sampling F and Cr values in the JADE algorithm it uses a memory to store the successful control parameters values.

The performance of DE algorithms highly depends on their search strategy [65,66]; therefore, some DE algorithms such as SaDE [67], CoDE [68], EPSDE [69], and MPEDE [70] have been introduced by combining different search strategies. In these algorithms, the search strategies are applied to each individual of the population in order to find the survived individuals for the next iteration. Therefore, using different search strategies can cover the weak points of each other in solving various complex optimization problems. In DE choosing search strategies and their associated control parameter values is imperative in performance improvement. Depending on the circumstances, different strategies and parameter settings are essential to increase the performance. Therefore, a combination of multiple search strategies of DE has been developed, which can affect the performance of DE. Thus, it is our motivation to propose a DE variant algorithm enriched by an efficient combination of search strategies, in which both search strategies and their associated control parameter values are gradually learned from their previous experiences in generating promising solutions.

In this work, firstly, a new multi-trial vector approach called MTV, is proposed to provide a meaningful search strategy by using different trial vector producers (TVPs), which cover a variety of problems properties. Another advantage of MTV is the opportunity of defining a winner-based distribution policy in order to distribute the population between TVPs. It also provides a life-time archive to preserve and share the information of recently promising solutions. Secondly, we propose an effective differential evolution algorithm named multi-trial vector-based differential evolution algorithm or MTDE. It uses the MTV approach by three different TVPs: representative based trial vector producer (R-TVP), local random based trial vector producer (L-TVP), and global best history based trial vector producer (G-TVP). These TVPs are introduced to solve different and complex problems by enriching the search strategy such that R-TVP maintains the diversity, L-TVP benefits from a proper balance between exploration and exploitation, and G-TVP represents a major ability in exploitation. Moreover, in MTDE, the winner-based distributing and its policy are defined based on behaviors of R-TVP, L-TVP, and G-TVP to distribute the population between them.

To summarize, the differences of this study compared to the existing approaches are as follows. The previous works distribute the main population into some smaller subpopulations that have the same sizes, but relying on the defined distribution policy, the sizes of subpopulations in this study are not equal. Considering a life-time archive makes the MTV approach able to maintain the population diversities while the evolution of the individuals. The most of the related works use the same search strategy in different subpopulations, while the MTDE is enriched by three mutation strategies. Moreover, in the MTDE, the sizes of subpopulations in this study are not equal, and the best search strategy will be rewarded with larger population allocation, while the worst search strategy will be penalized.

The rest of this article is organized as follows. The MTV approach and the MTDE algorithm are described in Sections 2 and 3. The experimental and statistical results are presented and discussed in Section 4. The applicability of the MTDE for solving real application problems is tested by engineering problems in Section 5. Section 6 discusses the main reasons for successes of the MTDE and its search strategies. Finally, Section 7 concludes the work and suggests future directions.

2. Multi-trial vector (MTV) approach

As per the No-free-lunch theorem [71], there will not be a general optimization algorithm superior to all others. In other words, the search strategy used in each optimization algorithm has its respective advantages in dealing with particular optimization problems. Thus, using a combination of search strategies can be an effective way to cover a variety of problems which is our motivation for using multiple search strategies instead of only one search strategy. Therefore, this section is to propose a new approach named multi-trial vector (MTV), which benefits from several TVPs for solving different problems. Each TVP has its specific approach to produce trial vectors as candidates for new positions of the existing individuals in the search space. As shown in Fig. 2, the MTV approach consists of four phases: winner-based distributing, multi-trial vector producing, evaluating and population updating, and life-time archiving. Each of these phases are briefly explained in the following paragraphs.

Winner-based distributing phase: In the MTV approach, the number of iterations is divided into k portions named *WinIter*, each including n iterations. In each iteration, the population is randomly distributed between TVPs using a distribution policy, which is determined based on the winner TVP. The winner TVP is a TVP with the most improvement in the past *WinIter* iterations

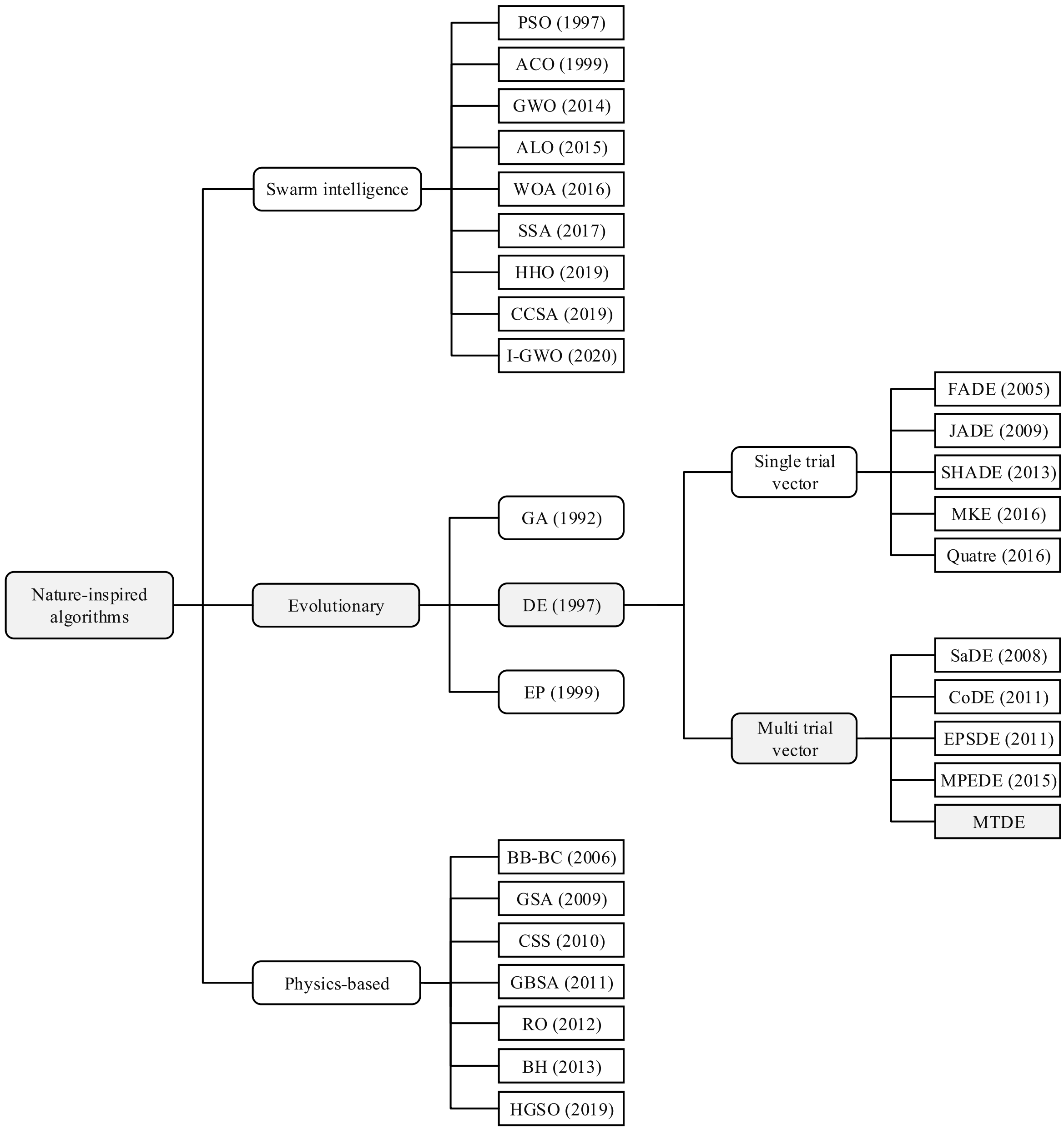

Fig. 1. A classification of the nature-inspired metaheuristic algorithms.

passed by the dashed line in Fig. 2. In this model, the size of *WinIter* is a trade-off, and it can be fine-tuned to strike a balance between exploitation and exploration.

Multi-trial vector producing phase: This phase is the movement part of the MTV approach provided by a combination of different TVPs. In each *WinIter*, the number of improvement obtained by each TVP is calculated to determine the winner TVP and the distribution policy for the next *WinIter*. The distribution policy can be mixed by reward or penalty regarding the behavior of the winner TVP. There is a kind of implicit information sharing in the MTV approach when the winner TVP is changed, and then the individuals are exchanged between subpopulations. Also, TVPs can use other individuals from out of their subpopulations, such as an archived population.

Evaluating and population updating phase: In this phase, the trial vectors produced by each TVP are evaluated to update the population by those trial vectors that could be improved. If the fitness value of the produced trial vector is less than its corresponding individual, then the position of the individual is updated by the produced trial vector's position. Otherwise, the individual remains unchanged in the population.

Life-time archiving phase: Since those individuals that have been changed by their produced trial vectors contain some information about the promising areas that recently explored, their information can be useful for sharing, especially in the last iterations. Therefore, these inferior solutions are archived in this phase. To evaluate the value of these archived solutions, their age is kept by a lifetime variable, which increases by one for each iteration. In the beginning, the archive is initiated empty, then the previous position of updated individuals is added to the archive in the course of iterations. If the size of the archive exceeds a predefined size, then some archived individuals can be removed from the archive using a suitable policy, e.g. using their age. Using this life-time archiving can improve the diversity and prohibit premature convergence of the MTV approach.

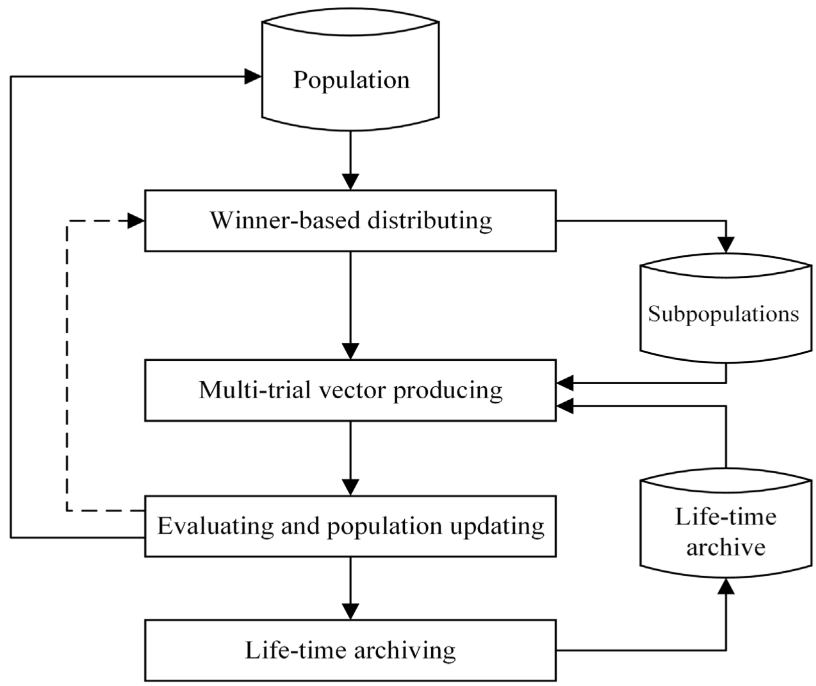

Fig. 2. The model of multi-trial vector (MTV) approach.

3. Multi-trial vector-based differential evolution (MTDE) algorithm

This section is to explain how the proposed MTDE is designed and implemented using the MTV approach. The importance of its design is to address the no free lunch theorem in which there is a need for different search strategies to solve different optimization problems. No single algorithm can show superior performance for all global optimization problems with different characteristics. In other words, depending on several factors, such as the multi-modality of the problem, the chosen EA and global exploration/local exploitation stages of the search process, using different search strategies can be effective on different problems and during different stages of the search process. As the flowchart shown in Fig. 3, the proposed MTDE algorithm includes two main steps: initializing and movement, which is based on the MTV approach. In the initializing step, N individuals $x_1, \ldots, x_N$ are randomly distributed in the search space by considering a restricted boundary between lower bound (l) and upper bound (u) using Eq. (1).

$$x_{ij} = l_j + \left(u_j - l_j\right) \times rand(0.1) \tag{1}$$

Where xij is the ith individual's position in jth dimension, l_j and u_j are the lower and upper bound of the jth dimension. The individual x_i is presented by vector $x_i = \{x_{i1}, x_{i2}, \ldots, x_{iD}\}$, where D denotes the number of dimensions of the search space, and N individuals are kept in a matrix $X_{N\times D}$. The fitness of each individual is calculated by the fitness function, $f(x_i)$, and the individual with minimum fitness value is considered as the gbest. The movement step is described in the following subsection in detail since it is the main part of the MTDE algorithm.

3.1. Movement step using MTV approach

The movement step of MTDE is based on the above-introduced MTV approach considering three trial vector producers: R-TVP, L-TVP, and G-TVP. Introducing the combination of these three trial vector producers is to provide MTDE for solving different problems. Particularly, R-TVP is to prevent local optima trapping and maintain the diversity, L-TVP benefits from a fast convergence and a proper balance between exploration and exploitation and G-TVP represents a major ability in exploitation and escape the local optima. As shown in Fig. 3, the movement step of the MTDE algorithm consists of four substeps: winner-based distributing, multi-trial vector producing, evaluating, and life-time archiving. The winner-based distributing substep is based on Definitions 1 and 2 as follows.

Definition 1. Given k *WinIter* each including n iterations, and only one winner TVP for each *WinIter* denoted by Win-TVP. In the first *WinIter*, R-TVP is considered as Win-TVP, and for the next *WinIters*, the Win-TVP is the TVP which has a higher improved rate in the past *WinIter*. The improved rate of X-TVP (X is R, L and G) denoted by $\mathrm{IR}_{\text{X-TVP}}$ is calculated by Eq. (2).

$$\mathrm{IR}_{\text{X-TVP}} = \frac{\mathrm{IF}_{\text{X-TVP}}}{\mathrm{FE}_{\text{X-TVP}}} \tag{2}$$

Where $\mathrm{IF}_{\text{X-TVP}}$ and $\mathrm{FE}_{\text{X-TVP}}$ are the number of improved finesses by the X-TVP and the number of function evaluations in a *WinIter*. After determining the Win-TVP, the winner-based distributing substep distributes the population between the trial vectors R-TVP, L-TVP, and G-TVP using the distribution policy based on Definition 2.

Definition 2. Given $N_{\text{R-TVP}}$, $N_{\text{L-TVP}}$ and $N_{\text{G-TVP}}$ are respectively the sizes of the subpopulations of $X_{\text{R-TVP}}$, $X_{\text{L-TVP}}$, and $X_{\text{G-TVP}}$. Then, N individuals are distributed randomly between these subpopulations, and their size is determined by two reward and penalty rules defined in Eqs. (3) and (4) based on the Win-TVP.

Reward rule: If R-TVP or L-TVP is Win-TVP, then

$$N_{\text{Win-TVP}} = 0.6 \times N \text{ and } N_{\text{Loser-TVPs}} = 0.2 \times N \tag{3}$$

Penalty rule: If G-TVP is Win-TVP, then

$$N_{\text{G-TVP}} = 0.2 \times N \text{ and } N_{\text{Loser-TVPs}} = 0.4 \times N \tag{4}$$

Before describing how three introduced TVPs apply their movement strategy, the following paragraphs provide some preliminaries and essential definitions. Both R-TVP and G-TVP cross

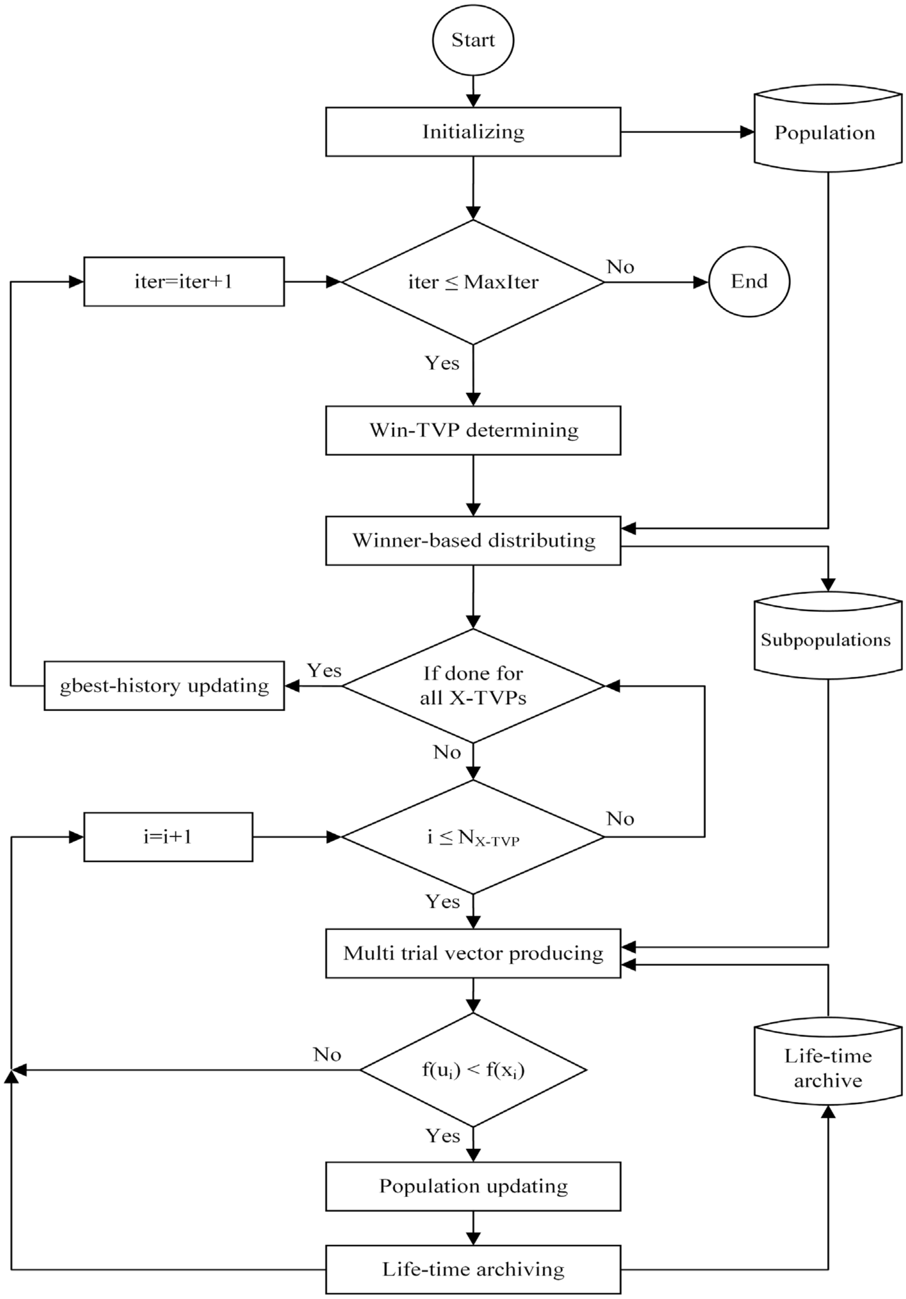

Fig. 3. The flowchart of MTDE algorithm.

their produced trial vectors over their corresponding individuals in order to form the final produced trial vectors. It is done by using a transformation matrix M and its binary inverse $\bar{M}$ which are created in each iteration. The matrix $M_{N\times D}$ is generated from $\tilde{M}_{D\times D}$, which is a lower triangular matrix with the elements set by one. First, $\tilde{M}$ is copied in M for (N/D) times, while if N cannot be divided by D and r is its remainder, then the last r rows of M are filled by the first r lines of $\tilde{M}$. Next, each row vector of M and then their order are randomly permuted. Finally, $\bar{M}$ is created by inverting the Boolean value of each element in M. Moreover, R-TVP and L-TVP use an additional population named unionPopulation ($X_{\text{U-pop}}$), which is the union of the individuals of the current population and the lifetime archive in order to maintain diversity and information sharing.

For each individual x_i, a scale factor F [63] is calculated by Cauchy distribution such that $F_i = randc_i\ (\mu f,\ \sigma)$, where μ_f is the mean value of improved scale factors and σ is the variance with constant value 0.2. The value of F_i must be between (0, 1], then if $F_i > 1$, it is considered by 1, and if $F_i \leq 0$ it will be calculated again, also the parameter μ_f is initiated to 0.5. For other iterations, if there is any improved individual in the population, μ_f is calculated by weighted Lehmer mean shown in Eq. (5), and if there is no improved individual, the scale factors remain unchanged.

$$\mu_f = \frac{\sum_{f_i\in S_f} w_{f_i} \times f_i^2}{\sum_{f_i\in S_f} w_{f_i} \times f_i} \tag{5}$$

Where S_f is the set of all Fs which $f(u_i) < f(x_i)$ in the current iteration and the weight w_{f_i} is calculated as given in Eq. (6), where

$\Delta f_i = f(x_i) - f(u_i)$.

$$w_{f_i} = \frac{\Delta f_i}{\sum_{f_i \in S_f} \Delta f_i} \tag{6}$$

Since the R-TVP is designed to help with escaping local optima and maintaining the diversity, it moves individual x_i from its subpopulation using positions of a random individual x_{u_pop} from the unionPopulation and two individuals, x_{i_best} and x_{i_worst}. The x_{i_best} and x_{i_worst} are respectively the ith elements of the ascending and descending ordered representation of the fitness values of the subpopulation $X_{\text{R-TVP}}$. Then, the trial vector u_i in R-TVP is produced by Eq. (7) using the M and $\bar{M}$ matrixes and v_i computed by Eq. (8).

$$u_i = M \times x_i + \overline{M} \times v_i \tag{7}$$

$$v_i = x_i + F_i \times \left(x_{i_{best}} - x_i\right) + F_i \times \left(x_{i_{worst}} - x_i\right) + a_1 \times \left(x_{u\_pop} - x_i\right) \tag{8}$$

Where a_1 is calculated by Eq. (9), which is a linearly decreased coefficient [6] to apply higher values at early stages and smaller values of coefficient in the later stages of the search process. The pseudo-code of R-TVP shown in Fig. 4.

$$a_1 = 2 - iter \times (2/MaxIter) \tag{9}$$

L-TVP benefits from a fast convergence and a proper balance between exploration and exploitation by learning from not only the individuals of its subpopulation but also other individuals from the unionPopulation. It changes the position of x_i by considering the difference between positions of two randomly selected individuals x_{r1} and x_{r2} from $X_{\text{L-TVP}}$ and the difference of x_i from a randomly selected individual x_{u_pop} derive from the unionPopulation as shown in Eq. (10).

$$v_i = x_i + F_i \times (x_{r1} - x_{r2}) + a_2 \times \left(x_{u\_pop} - x_i\right) \tag{10}$$

Where a_2 is a non-linear decreased coefficient [72] which calculated by Eq. (11) as follows:

$$a_2 = initial - (initial - final) \times (((MaxIter - iter)/MaxIter))^{Mu} \tag{11}$$

Where the *initial* and *final* are the initial and final values of control parameter a_2 and *Mu* is a dimension dependent value. Differ from the R-TVP, the final trial vector u_i is based on individual learning, and not an evolution, so it does not use a crossover and $u_i = v_i$. The pseudo-code of L-TVP shown in Fig. 5.

G-TVP is introduced to overcome the shortcoming of trapping in the local optima by using a history of *gbest* named *gbest-history*. The gbest-history is designed in a way to keep H numbers of the best-obtained individuals so far, which enables the G-TVP to learn from multiple different gbests. This memory stores and shares information about the high quality explored regions of the search space and already found good solutions. In the first iteration, gbest-history is initialized with only the gbest. Then, in the next iterations, the gbest of each iteration is added to the gbest-history distinctly. When gbest-history is full, the current gbest is replaced with the existed gbest in the history with higher fitness value.

Also, for the G-TVP, a matrix $M_{\text{gb-h}}$ which has $N_{\text{G-TVP}}$ rows and D columns, is constructed from the gbest-history. This construction can prevent the loss of diversity by using different best individuals. In each iteration, the $M_{\text{gb-h}}$ is constructed by replicating the gbest-history for (N/m) times (m is the number of gbest-history members). Then, the trial vector u_i is produced by Eq. (12) similar in the R-TVP using the M and $\bar{M}$ matrixes and v_i computed by Eq. (13) where $x_{i\_\text{gb-h}}$ is the ith row of the $M_{\text{gb-h}}$, and x_{r1} and x_{r2} are two randomly selected individuals from the $X_{\text{G-TVP}}$. The pseudo-code of G-TVP shown in Fig. 6.

$$u_i = M \times x_i + \overline{M} \times v_i \tag{12}$$

$$v_i = x_{i\_gb-h} + a_2 \times (x_{r1} - x_{r2}) \tag{13}$$

In the evaluating substep, the quality of each trial vector u_i from $U_{\text{X-TVP}}$ is compared with its corresponding individual x_i from $X_{\text{X-TVP}}$. As shown in Fig. 3, if $f(u_i) \geq f(x_i)$, then x_i remains unchanged in the population. Otherwise, if it could be improved, i.e., $f(u_i) < f(x_i)$, the population updating procedure updates the position of x_i by u_i. Additionally, the number of individuals improved by each TVP is increased by one, and the value of F_i is stored for using in Eq. (5). Then, life-time archiving substep adds x_i to the life-time archive as an inferior solution. In the beginning, the life-time archive is empty, and it can keep the position and lifetime of N individuals. At the end of each iteration, the lifetime of archived individuals is increased by one, and N younger archived individuals with smaller lifetime are maintained.

Finally, the counter of iterations ($iter$) is increased by one, and the evolutionary search can be iterated until the predefined number of iterations ($MaxIter$) is reached. The pseudo-code of the proposed MTDE algorithm shown in Fig. 7.

3.2. Computational complexity of MTDE

As Fig. 6 shown, the MTDE algorithm has two main steps initializing and movement. The computational complexity of the initializing step for distributing N individuals in the search space with D dimensions is O(ND). In the movement step, first, the computational complexity of the winner-based distributing substep (line 10) is O(N). Next, the computational complexity of Do-loop (lines 11–21) including multi-trial vector producing and evaluating substeps is:

O $(2(N_{\text{R-TVP}} \log N_{\text{R-TVP}}) + N_{\text{R-TVP}}D + N_{\text{L-TVP}}D + N_{\text{G-TVP}}D + ND)$ where the computational cost $2(N_{\text{R-TVP}} \log N_{\text{R-TVP}})$ is for the ascending and descending ordered representation of the fitness values in R-TVP with $N_{\text{R-TVP}}$ individuals. Because $N = N_{\text{R-TVP}} + N_{\text{L-TVP}} + N_{\text{G-TVP}}$, then the computational complexity of this Do-loop is $O(2(ND + N_{\text{R-TVP}} \log N_{\text{R-TVP}}))$. The cost of constructing matrix $M_{\text{gb-h}}$ (line 24) is O(N), then computational complexity of the movement step for all iterations (T) is $O(T(2(N + ND + N_{\text{R-TVP}} \log N_{\text{R-TVP}})))$ which results the overall computational complexity of the MTDE algorithm is equal to O(ND+2TND) or O(TND).

4. Experimental evaluation of MTDE algorithm

To evaluate the performance of the MTDE algorithm, a variety of experiments is performed on several benchmark functions. First, a qualitative analysis of our proposed algorithm is provided by a set of different experiments on four metrics: search history, trajectory, average fitness values, and gbest values. The aim of these experiments is to show the exploratory and exploitative abilities of MTDE and its convergence behavior. Second, a quantitative analysis including several experiments is performed to evaluate the performance of the MTDE algorithm by comparing with the state-of-the-art metaheuristic algorithms.

4.1. Experimental environment and benchmark functions

The MTDE and comparative algorithms were implemented in Matlab R2018a, and all experiments were run on a CPU, Intel Core(TM) i7-3770 3.4 GHz, and 8.00 GB RAM. We used the benchmark functions CEC 2018 [73] as it provides a variety of challenging test functions consists of four groups: unimodal, multimodal, hybrid, and composition functions. Functions F_1 and F_3 are unimodal with one optimum, so they are suitable for benchmarking the exploitation ability and convergence analysis of the proposed algorithm. Functions F_4 - F_{10} are multimodal functions with many local optima that are particularly used to test the exploration ability and local optima avoidance of MTDE.

Algorithm 1: Representative based trial vector producer (R-TVP)

Input: $X_{R\text{-}TVP}$, $X_{U\text{-}pop}$, M, $\bar{M}$

Output: $U_{R\text{-}TVP}$ (produced trial vectors)

Procedure R-TVP
  for i = 1 to $N_{R\text{-}TVP}$
    $v_i = x_i + F_i * (x_{i_{best}} - x_i) + F_i * (x_{i_{worst}} - x_i) + a_1 * (x_{u\_pop} - x_i)$
    $u_i = M \times x_i + \bar{M} \times v_i$
    $U_{R-TVP} = U_{R-TVP} + u_i$
  end
  Return produced vectors in $U_{R\text{-}TVP}$
End procedure

Fig. 4. Pseudo-code of R-TVP.

Algorithm 2: Local random based trial vector producer (L-TVP)

Input: $X_{L\text{-}TVP}$, $X_{U\text{-}pop}$

Output: $U_{L\text{-}TVP}$ (produced trial vectors)

Procedure L-TVP
  for i = 1 to $N_{L\text{-}TVP}$
    $v_i = x_i + F_i * (x_{r1} - x_{r2}) + a_2 * (x_{u\_pop} - x_i)$
    $u_i = v_i$
    $U_{L-TVP} = U_{L-TVP} + u_i$
  end
  Return produced vectors in $U_{L\text{-}TVP}$
End procedure

Fig. 5. Pseudo-code of L-TVP.

Algorithm 3: Global best history based trial vector producer (G-TVP)

Input: $X_{G\text{-}TVP}$, $M_{gb\text{-}h}$, M, $\bar{M}$

Output: $U_{G\text{-}TVP}$ (produced trial vectors)

Procedure G-TVP
  for i = 1 to $N_{G\text{-}TVP}$
    $v_i = x_{i\_gb-h} + a_2 * (x_{r1} - x_{r2})$
    $u_i = M \times x_i + \bar{M} \times v_i$
    $U_{G-TVP} = U_{G-TVP} + u_i$
  end
  Return produced vectors in $U_{G\text{-}TVP}$
End procedure

Fig. 6. Pseudo-code of G-TVP.

Functions F_{11} - F_{20} and F_{21} - F_{30} are hybrid and composition functions that are more complex and challenging than unimodal and multimodal functions. They are suitable for testing the potential of the algorithms in solving real-world problems and the ability of the MTDE algorithm in maintaining the balance between the exploration and exploitation.

4.2. Qualitative analysis

The qualitative analysis of the MTDE algorithm was performed by 100 individuals over 200 iterations on some selected functions from each group of the benchmark functions CEC 2018 with 10 decision variables. In this experiment set, we used four metrics: search history, trajectory, the average fitness values, and the gbest values of all individuals. Although the search history shows the sampled regions of the search space, it cannot show the order of exploration and exploitation performed during the course of iterations. Therefore, the trajectory metric is used to indicate the search strategy's behaviors over the search space, such as sudden changes in the initial iterations, which is for exploration and uniform behavior in the final steps for exploitation. Then, the average fitness values of all individuals in each iteration can show the ability of the MTDE algorithm in order to improve the solutions during the course of iterations. Finally, the gbest values are used to reveal the varying of the best solution during the optimization, and they can show the convergence ability of the MTDE algorithm. It is expected that a competitive metaheuristic algorithm converges to promising regions in the initial iterations, and the vicinity of the global optimum in the final iterations.

In the first column of Fig. 8, the black sampled points are the position history of all individuals in the search history during the optimization, and the global optimum solution is marked with a bigger red point. As per the search histories shown in the first column, the individuals are distributed sparsely in the unpromising regions. The coverage of a large portion of the search space is reasonable. Also, most of the individuals are gathered around the global optimum solution. Tracking of the individuals demonstrates the exploration ability of MTDE to find the promising areas and its exploitation ability to find the global optimum. In contrast to unimodal functions, multimodal functions have many local optima with the number increasing by growing the dimension. Thus, according to the search history of multimodal

```
Algorithm 4: The MTDE algorithm
Input: N, D, MaxIter, WinIter, H (gbest-history size)
Output: The global optimum (gbest)
Begin
iter = 1 and Win-TVP = R-TVP.
Randomly distribute N individuals in the search space.
Evaluating the fitness f(x_i) and set the gbest.
Constructing the M_gb-h matrix.
While iter ≤ MaxIter
    If mod (iter, WinIter) == 0
        Determining Win-TVP using Definition 1.
    End if
    Winner-based distributing (Win-TVP) using Definition 2.
    Do for each TVP
        For i = 1 to N_X-TVP
            Calculate F_i.
            Multi-trial vector producing.
            If f(u_i) < f(x_i)
                Updating x_i by u_i.
                Improved individual counting.
                Life-time archiving.
            End if
        End for
    End do
    Updating gbest.
    gbest-history updating.
    Constructing the M_gb-h matrix.
iter = iter + 1.
End while
Return the global optimum (gbest).
End
```

Fig. 7. Pseudo-code of the MTDE algorithm.

test function F_7 shown in the second row of Fig. 8, the MTDE algorithm is able to avoid the local optima as well.

The last four rows of Fig. 8 show the search history of hybrid and composition functions F_{14}, F_{18}, F_{26}, and F_{28} that demonstrate exploitation behavior similar as was discussed above. It is concluded that the introduced movement strategy is able to find an optimum solution in different groups of test functions.

The second column of Fig. 8 presents the trajectory of the first dimension of the first individual in the population. The trajectories of all functions shown in Fig. 8 present the abrupt changes in the individual movements in the initial iterations and the gradual changes in the last iterations. According to Berg [74], this behavior guarantees that our proposed algorithm eventually converges to a point in a search space.

The average fitness and the gbest values of all individuals are also visualized in the third and fourth columns of Fig. 8 in order to show the ability of the proposed algorithm for improving the fitness of initial individuals over the course of iterations. The MTDE algorithm quickly converges towards a good solution in the initial iterations, and there is a significant drop in the fitness value. Eventually, the convergence has a gradual decrease in the last iterations. The results shown in these two columns present that the MTDE algorithm has a fast convergence toward the optimum, and it is able to reach the global optimum.

4.3. Quantitative analysis

In this section, the performance of the proposed algorithm is evaluated and compared with the state-of-the-art metaheuristic algorithms: GWO [6], WOA [31], SSA [32], HHO [7], CoDE [68], EPSDE [69], QUATRE [59], and MKE [58]. Different experiments were conducted on 29 test functions with a varying dimension (D) 10, 30, and 50 to evaluate the exploration and exploitation, local optima avoidance, and convergence behavior of the MTDE algorithm. In all experiments, the population size N was 100, and the maximum number of iterations was equal with $MaxFEs/N$ where the maximum function evaluations ($MaxFEs$) were set to 10000*D. In each experiment, all algorithms were run 20 times, and results are reported based on the fitness error, $\Delta f = f - f^*$ where f is the optimization result obtained by the corresponding algorithm, and f^* is the global optimum. The mean, standard deviation, and minimum value of fitness error are employed to measure the performance of the algorithms.

The parameters settings are presented in Table 1 in which the comparative algorithms were set as same as the recommended settings in their original works. The experimental results shown in Tables 2–5 in which the bold values show the winning algorithms or the best solutions. At the end of each table, the comparison of the results by the numbers of win (w), tie (t) and loss (l) of each algorithm is shown.

4.3.1. Exploration and exploitation evaluation

As discussed in the preceding section, the unimodal functions are suitable for assessing the exploitation, whereas the multimodal functions can evaluate the exploration. Thus, the exploitation and exploration of MTDE were evaluated and compared with comparative algorithms by these two groups of benchmark functions as follows.

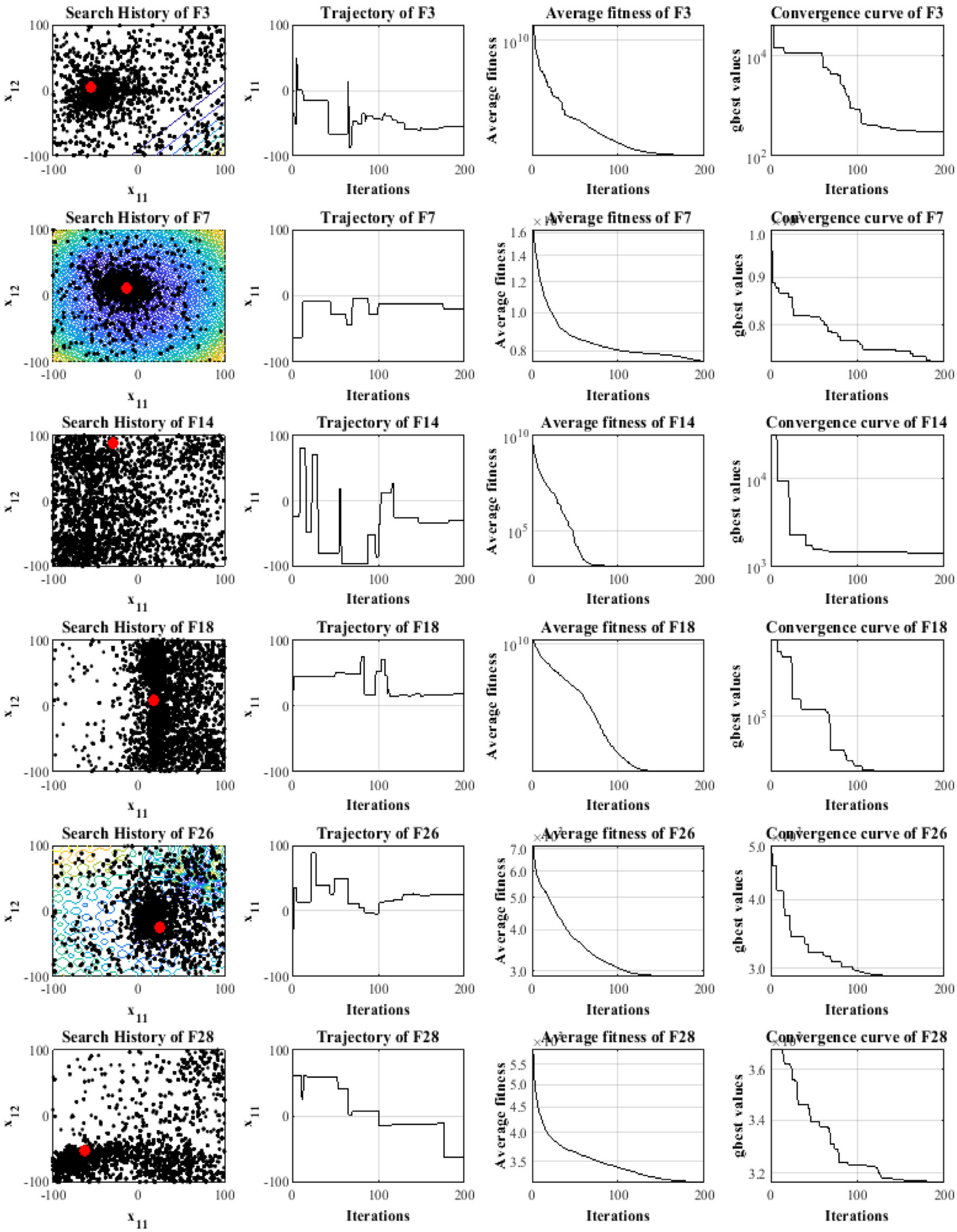

Fig. 8. Search history, trajectory in first dimension, average fitness, and convergence curve of MTDE.

As results reported in Table 2, the MTDE algorithm provides very competitive results for functions F_1 and F_3, especially in 10D and 30D problems. It can be stated that the MTDE algorithm converges towards the global optimum solution and exploit near it accurately. Meanwhile, the results shown in Table 3 prove the proposed algorithm is better than comparative algorithms for solving the benchmark functions F_4 - F_{10}, which contain many local optima.

4.3.2. Evaluation of local optima avoidance

Hybrid and composition functions are more complex and challenging than the other functions since their search landscape is made of multiple test functions. Thus, these are used to test the ability of MTDE to maintain a proper balance between exploration and exploitation that results in avoiding local optima.

Table 4 shows the results of solving hybrid functions by MTDE and other algorithms, where MTDE is superior to other algorithms, especially for 10D and 30D problems. Furthermore, the results in Table 5 provide evidence of the competitive performance

Table 1
Parameters settings.

Alg.	Setting
GWO	a = [2 0]
WOA	a = [2 0], r = [−1 1]
SSA	c_2 = rand [0 1], c_3 = rand [0 1]
CoDE	CR = [0.1 0.9 0.2], F = [1 1 0.8]
EPSDE	CR ∈ [0.1 0.9], F ∈ [0.4,0.9]
QUATRE	c = 0.7
MKE	FC = 0.7
MTDE	WinIter = 20, H = 5, initial = 0.001, final = 2, Mu = log(D), μf = 0.5, σ = 0.2

of MTDE for composition functions. Therefore, it is concluded that the proposed algorithm strikes a balance between exploration and exploitation to approximate the global optimum solution.

4.3.3. Convergence evaluation

In this experiment set, the ability and convergence speed of MTDE for improving the candidate or the gbest solution over the course of the iterations is evaluated and compared with the comparative algorithms. Figs. 9 and 10 show the comparison of the convergence curves of MTDE and comparative algorithms for some of the selected functions with different dimensions 10, 30, and 50. For drawing these curves for each algorithm, the mean of its gbest values over 20 runs was considered in each iteration.

Inspecting the convergence curves, it is evident that MTDE has three dissimilar convergence behaviors during the optimization process for solving different benchmark functions. First, as shown in Figs. 9 and 10, for 10D (F_1, F_3, F_{12}, F_{16}, F_{28}, F_{30}), 30D (F_3, F_5, F_6, F_8, F_{16}, F_{22}, F_{30}) and 50D (F_5, F_7, F_8, F_{10}, F_{20}, F_{21}, F_{22}) MTDE has an accelerated decreasing convergence in the initial iterations until the half of iterations where it reaches the optimum solution. The second behavior is the rapid convergence until the half of iterations, and the approximation of the global optimum becomes more accurate as iteration is increased which is evident on functions 10D ($F_{5,}$ F_8, F_{21}, F_{23}, F_{29}), 30D (F_{10}), and 50D (F_{12}, F_{18}). Finally, the last behavior registered for 10D (F_{10}), 30D ($F_{12,}$ F_{26}), and 50D (F_1, F_{12}) which is continuing to improve the solution until the final iterations. These results prove that the convergence of MTDE is competitive with other algorithms for various benchmark functions with different dimensions.

Table 6 summarizes all performance results of MTDE and other algorithms shown in Tables 2–5 by a useful metric named overall effectiveness (OE). The OE of each algorithm is computed by Eq. (14), where N is the total number of tests, and L is the total number of losing tests for each algorithm.

$$\textit{Overall Effectiveness}\ (OE) = ((N - L)/N) \times 100 \tag{14}$$

4.4. Impact analysis and comparison of using the TVPs

In this section, the impact of using the proposed search strategies L-TVP, R-TVP, and G-TVP on the performance of the MTDE algorithm is analyzed and compared by two different experiments. The results of these experiments shown in Figs. 11 and 12 on some selected functions from different categories of CEC 2018.

In the first experiment, Fig. 11 shows the best fitness obtained by all individuals in each iteration for each TVP uniquely and their combination, which is MTDE. As the curves show in this figure, the solutions obtained by G-TVP are better than other TVPs for unimodal function F_3 and the multimodal functions F_7, F_8, and F_{10}, which show the ability of G-TVP in the exploitation. Moreover, for the optimization of multimodal functions which contain many local optima, it can prevent the local optima trapping. The main reason is that using of the gbest-history helps to prevent the trapping in local optima and losing diversity. The L-TVP search strategy also can find better solutions for the hybrid functions F_{12}, F_{15}, and F_{18} and for the composition function F_{26}, F_{28}, and F_{30}. In these functions, L-TVP can make a proper balance between local and global search and avoid premature convergence. Similarly, the R-TVP has the same performance in some other hybrid and composition functions F_{15}, F_{22}, F_{30}, and F_{28}.

In the second experiment of this section, the improved rate of TVPs is evaluated and the percent of their improvement shown in Fig. 12. This figure reveals that the R-TVP has the best performance at the start, but the L-TVP and G-TVP show more impact than the R-TVP rapidly. The proportion of each TVP in improving the obtained solutions varies with different problems or even different stages of a problem. Although the experiment results in this section prove that the proposed TVPs can find the solutions individually, using all of TVPs has more benefits since the solutions obtained by MTDE is always better.

Table 2
The comparison of obtained solutions for unimodal functions.

F	D	Index	GWO	WOA	SSA	HHO	CoDE	EPSDE	QUATRE	MKE	MTDE
F1	10	Mean	4.3804E+06	2.3203E+05	2.3029E+03	2.2324E+05	1.2450E+00	**0.0000E+00**	**0.0000E+00**	**0.0000E+00**	**0.0000E+00**
		STD	1.1538E+07	4.3811E+05	2.4192E+03	1.0684E+05	7.3842E−01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
		Min	6.5544E+03	1.3644E+04	1.8116E+01	5.1910E+04	3.8841E−01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
	30	Mean	8.4874E+08	3.2594E+06	2.4745E+03	7.7264E+06	3.2904E+01	**0.0000E+00**	**0.0000E+00**	**0.0000E+00**	**0.0000E+00**
		STD	7.7867E+08	2.3799E+06	3.8492E+03	2.0976E+06	1.3274E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
		Min	4.1879E+07	7.7899E+05	3.0040E−02	4.3794E+06	1.8245E+01	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
	50	Mean	4.0637E+09	9.7205E+06	8.3558E+03	3.4005E+07	3.2764E+03	**0.0000E+00**	3.2867E−04	2.3984E−02	3.6737E−05
		STD	2.1433E+09	1.1581E+07	9.7953E+03	6.5147E+06	2.6506E+03	0.0000E+00	5.4499E−04	3.8457E−02	5.3479E−05
		Min	7.7500E+08	2.5276E+06	9.3768E+00	2.1808E+07	6.3653E+00	0.0000E+00	2.4435E−06	4.7969E−04	6.5947E−07
F3	10	Mean	5.0896E+02	2.1389E+02	1.4429E−09	7.6513E−01	1.4151E−02	**0.0000E+00**	**0.0000E+00**	2.0407E−06	**0.0000E+00**
		STD	1.9473E+03	2.5377E+02	5.7929E−10	3.5665E−01	6.8943E−03	0.0000E+00	0.0000E+00	3.8251E−06	0.0000E+00
		Min	2.0144E−01	1.2343E+01	6.6109E−10	2.2879E−01	3.7083E−03	0.0000E+00	0.0000E+00	1.7563E−08	0.0000E+00
	30	Mean	2.8332E+04	1.6691E+05	4.3359E−08	5.4254E+02	9.9455E−01	2.1993E+04	1.1429E−03	6.8870E+03	**0.0000E+00**
		STD	8.0961E+03	7.7951E+04	1.3045E−08	3.3973E+02	5.4310E−01	6.8057E+04	1.8330E−03	2.4045E+03	0.0000E+00
		Min	1.1019E+04	4.9414E+04	2.1494E−08	2.1104E+02	3.0312E−01	0.0000E+00	4.4633E−05	3.5480E+03	0.0000E+00
	50	Mean	6.5679E+04	7.4951E+04	1.8672E−07	2.3721E+03	9.4758E+00	2.5566E+05	4.6298E+01	5.3530E+04	**1.5335E−07**
		STD	1.5390E+04	2.9689E+04	5.4193E−08	6.4084E+02	5.7952E+00	2.5029E+05	3.4788E+01	9.1203E+03	1.2034E−06
		Min	4.2100E+04	4.3995E+04	1.2759E−07	1.3515E+03	1.6755E+00	1.5037E+01	1.6302E+01	3.7412E+04	1.6787E−07
Ranking	**10**	(w/t/l)	0/0/2	0/0/2	0/0/2	0/0/2	0/0/2	0/2/0	0/2/0	0/1/1	0/2/0
	30	(w/t/l)	0/0/2	0/0/2	0/0/2	0/0/2	0/0/2	0/1/1	0/1/1	0/1/1	1/1/0
	50	(w/t/l)	0/0/2	0/0/2	0/0/2	0/0/2	0/0/2	1/0/1	0/0/2	0/0/2	1/0/1

Table 3
The comparison of obtained solutions for multimodal functions.

F	D	Index	GWO	WOA	SSA	HHO	CoDE	EPSDE	QUATRE	MKE	MTDE
F4	10	Mean	1.5966E+01	2.2290E+01	4.0763E+00	1.4983E+01	5.0253E−03	**0.0000E+00**	**0.0000E+00**	1.7375E−02	**0.0000E+00**
		STD	1.9469E+01	3.5023E+01	1.6623E+00	2.2726E+01	1.6899E−03	0.0000E+00	0.0000E+00	1.3101E−02	0.0000E+00
		Min	6.7447E+00	1.0902E+00	8.2176E−02	7.8072E−01	1.9805E−03	0.0000E+00	0.0000E+00	1.2041E−04	0.0000E+00
	30	Mean	1.4472E+02	1.5935E+02	9.1465E+01	1.1735E+02	8.1881E+01	**0.0000E+00**	5.1366E+01	3.0160E+01	1.4832E+01
		STD	3.1260E+01	4.2152E+01	1.8331E+01	3.6294E+01	6.2494E+00	0.0000E+00	2.1711E+01	3.0334E+01	1.8067E+01
		Min	9.3711E+01	7.8484E+01	5.9177E+01	3.7118E+01	6.2573E+01	0.0000E+00	2.8922E−06	1.4540E−04	8.9298E−06
	50	Mean	4.1183E+02	2.8050E+02	1.5207E+02	2.1326E+02	5.7725E+01	**1.1016E+01**	6.1393E+01	8.6500E+01	7.7638E+01
		STD	1.6721E+02	5.5343E+01	5.1341E+01	3.5729E+01	4.0601E+01	4.5064E+00	5.0933E+01	5.9118E+01	3.5456E+01
		Min	2.0800E+02	2.0982E+02	3.2501E+01	1.1932E+02	2.8529E+01	2.2593E−01	2.6546E−01	3.3654E−05	1.8570E+01
F5	10	Mean	1.3829E+01	4.8430E+01	1.9530E+01	4.1438E+01	1.3221E+01	4.8495E+00	7.8969E+00	1.0355E+01	**3.8078E+00**
		STD	8.1265E+00	1.4771E+01	6.6982E+00	1.3220E+01	2.2504E+00	1.1777E+00	2.9965E+00	2.0532E+00	1.0651E+00
		Min	2.9900E+00	2.3914E+01	8.9546E+00	2.1042E+01	9.0361E+00	2.6152E+00	4.9748E+00	6.1365E+00	1.9900E+00
	30	Mean	9.1704E+01	2.6365E+02	1.3142E+02	2.1057E+02	1.3542E+02	5.5080E+01	6.2296E+01	1.1132E+02	**2.1093E+01**
		STD	3.3006E+01	5.4562E+01	4.2581E+01	3.0208E+01	1.0637E+01	5.7163E+00	1.7996E+01	1.0193E+01	4.4683E+00
		Min	4.7122E+01	1.7902E+02	6.4672E+01	1.5774E+02	1.1191E+02	4.1085E+01	3.7808E+01	7.7546E+01	1.1940E+01
	50	Mean	1.8244E+02	4.0646E+02	2.9456E+02	3.6692E+02	2.9809E+02	1.7615E+02	1.5493E+02	2.4945E+02	**5.1141E+01**
		STD	3.7258E+01	7.6040E+01	7.1917E+01	3.6869E+01	1.6338E+01	1.4576E+01	3.2426E+01	1.4520E+01	7.5801E+00
		Min	1.2100E+02	2.7253E+02	1.7909E+02	2.7579E+02	2.7381E+02	1.3990E+02	9.2970E+01	2.1609E+02	3.6813E+01
F6	10	Mean	1.7018E−01	2.5601E+01	6.5687E+00	2.3223E+01	1.2604E−03	**0.0000E+00**	7.1056E−08	1.7764E−08	**0.0000E+00**
		STD	2.1016E−01	1.2741E+01	7.6390E+00	8.1746E+00	2.4123E−04	0.0000E+00	3.1777E−07	7.9443E−08	0.0000E+00
		Min	2.1790E−02	7.1904E+00	6.2463E−02	8.7868E+00	8.0572E−04	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
	30	Mean	4.0997E+00	6.6469E+01	3.5998E+01	5.8442E+01	3.4902E−02	**0.0000E+00**	5.8300E−06	6.1419E−08	1.9418E−05
		STD	1.8595E+00	8.9686E+00	1.4689E+01	7.0297E+00	6.8174E−03	0.0000E+00	8.1565E−06	2.4437E−07	1.3975E−05
		Min	1.1431E+00	5.0184E+01	1.4778E+01	4.8136E+01	2.4335E−02	0.0000E+00	1.1369E−13	1.1369E−13	1.5934E−06
	50	Mean	1.0505E+01	7.6778E+01	4.3640E+01	6.8191E+01	2.0724E−02	**0.0000E+00**	4.1596E−02	9.7842E−07	6.6010E−04
		STD	3.9121E+00	9.8048E+00	8.4093E+00	4.8142E+00	5.2702E−03	0.0000E+00	1.5616E−01	2.0840E−06	6.4921E−04
		Min	5.5000E+00	6.5790E+01	2.4037E+01	5.8976E+01	1.1827E−02	0.0000E+00	1.5023E−04	2.2737E−13	1.1487E−04
F7	10	Mean	2.6981E+01	7.4922E+01	3.4818E+01	6.6597E+01	2.9838E+01	1.5672E+01	1.8095E+01	2.1450E+01	**1.2325E+01**
		STD	7.0974E+00	2.3504E+01	1.0905E+01	2.0692E+01	3.0239E+00	1.3268E+00	4.7174E+00	3.6063E+00	7.9495E−01
		Min	1.4216E+01	3.5879E+01	1.7532E+01	2.5252E+01	2.3124E+01	1.3595E+01	3.1395E+00	1.3001E+01	1.0367E+01
	30	Mean	1.2569E+02	4.8959E+02	1.4110E+02	5.0835E+02	1.7911E+02	8.8058E+01	1.0406E+02	1.4987E+02	**5.4444E+01**
		STD	3.1442E+01	1.0447E+02	3.5455E+01	8.4135E+01	9.3412E+00	7.0595E+00	1.6057E+01	1.3095E+01	4.9194E+00
		Min	7.1452E+01	3.3673E+02	9.2872E+01	3.0118E+02	1.5585E+02	7.5753E+01	7.7892E+01	1.1808E+02	3.9650E+01
	50	Mean	3.3471E+02	9.9487E+02	3.3166E+02	1.0640E+03	3.7033E+02	2.3271E+02	2.2284E+02	3.1354E+02	**1.0853E+02**
		STD	8.5908E+01	7.3579E+01	9.5849E+01	9.9832E+01	1.1546E+01	1.2277E+01	2.5890E+01	1.4726E+01	9.7647E+00
		Min	2.0700E+02	7.6593E+02	2.1849E+02	7.5078E+02	3.4773E+02	2.0152E+02	1.5672E+02	2.9048E+02	9.0899E+01
F8	10	Mean	1.2545E+01	4.0125E+01	2.4526E+01	2.6711E+01	1.5228E+01	5.0498E+00	1.0298E+01	1.0633E+01	**3.1373E+00**
		STD	6.2583E+00	2.1323E+01	1.3674E+01	6.4088E+00	2.9700E+00	1.2730E+00	3.7675E+00	2.2044E+00	8.7017E−01
		Min	4.3893E+00	1.5924E+01	3.9798E+00	1.4993E+01	8.7496E+00	1.9541E+00	2.9849E+00	7.5543E+00	9.9496E−01
	30	Mean	7.6365E+01	2.0831E+02	1.2643E+02	1.4503E+02	1.3819E+02	5.7636E+01	7.3391E+01	1.1535E+02	**2.6217E+01**
		STD	1.4647E+01	4.7504E+01	3.3405E+01	1.2598E+01	9.6697E+00	6.5712E+00	1.8403E+01	7.5100E+00	5.9196E+00
		Min	5.7489E+01	1.2754E+02	8.0660E+01	1.2330E+02	1.1184E+02	4.6001E+01	3.3445E+01	9.8898E+01	1.2934E+01
	50	Mean	1.9575E+02	4.1536E+02	2.8910E+02	3.4872E+02	2.9348E+02	1.7623E+02	1.6519E+02	2.4837E+02	**4.9698E+01**
		STD	3.1627E+01	7.3544E+01	8.7490E+01	2.5808E+01	1.9818E+01	1.1684E+01	3.5980E+01	1.5700E+01	7.2576E+00
		Min	1.3600E+02	2.9331E+02	1.8108E+02	2.8831E+02	2.3861E+02	1.5395E+02	8.3315E+01	2.2429E+02	3.7808E+01
F9	10	Mean	2.2349E+00	4.7914E+02	1.9077E+00	3.8751E+02	1.6189E−05	**0.0000E+00**	**0.0000E+00**	**0.0000E+00**	**0.0000E+00**
		STD	4.2107E+00	3.5934E+02	5.7342E+00	2.2254E+02	9.4832E−06	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
		Min	5.3691E−02	8.1739E+01	1.0925E−10	1.1517E+02	5.4588E−06	0.0000E+00	0.0000E+00	0.0000E+00	0.0000E+00
	30	Mean	5.4051E+02	5.9263E+03	2.7962E+03	5.2283E+03	5.2404E−01	**3.2313E−03**	5.4966E−01	9.5341E−02	1.9482E−01
		STD	4.4070E+02	1.9896E+03	1.3262E+03	7.0020E+02	2.2753E−01	1.4451E−02	1.4451E+00	1.8522E−01	2.0698E−01
		Min	9.5388E+01	2.6087E+03	4.7416E+02	3.8062E+03	1.9247E−01	0.0000E+00	0.0000E+00	1.1369E−13	0.0000E+00
	50	Mean	4.5701E+03	1.9529E+04	9.4717E+03	1.5619E+04	**6.2810E−02**	9.6120E−01	1.0404E+02	5.0356E−01	7.0265E+00
		STD	3.2583E+03	4.9687E+03	2.6604E+03	2.2633E+03	9.8386E−02	2.8788E+00	3.3006E+02	5.0871E−01	3.0468E+00
		Min	1.5100E+03	1.1229E+04	3.5428E+03	1.2553E+04	6.4343E−03	0.0000E+00	2.6858E−01	2.2737E−13	2.0859E+00
F10	10	Mean	5.3932E+02	1.0556E+03	7.3398E+02	8.8166E+02	6.1233E+02	4.1775E+02	4.0498E+02	5.0083E+02	**6.5005E+01**
		STD	1.9411E+02	2.9700E+02	2.4094E+02	2.6496E+02	8.4933E+01	1.1527E+02	2.4983E+02	1.1857E+02	6.0633E+01
		Min	6.8930E+01	6.0923E+02	2.2726E+02	4.5702E+02	4.3395E+02	1.1573E+02	9.7097E+00	2.5764E+02	3.8909E+00
	30	Mean	2.8508E+03	5.2298E+03	3.8357E+03	4.5525E+03	5.1066E+03	4.1857E+03	3.1701E+03	4.4401E+03	**2.2454E+03**
		STD	5.5476E+02	6.3974E+02	7.8230E+02	6.3532E+02	2.8878E+02	3.0389E+02	4.2558E+02	4.6790E+02	2.5885E+02
		Min	1.8596E+03	3.8322E+03	2.5933E+03	3.3727E+03	4.5302E+03	3.6896E+03	2.2016E+03	3.3142E+03	1.7433E+03
	50	Mean	5.6339E+03	9.0709E+03	6.6865E+03	7.4884E+03	1.0137E+04	9.3095E+03	7.1881E+03	9.2877E+03	**4.5467E+03**
		STD	6.9439E+02	1.3014E+03	9.6470E+02	9.9624E+02	3.3388E+02	4.1284E+02	7.3504E+02	4.6803E+02	6.0180E+02
		Min	4.2300E+03	7.5065E+03	4.8079E+03	6.0552E+03	9.5463E+03	8.7095E+03	4.1899E+03	7.8897E+03	3.2986E+03
Ranking	**10**	(w/t/l)	0/0/2	0/0/3	0/0/4	0/0/5	0/0/6	0/3/4	0/2/5	0/1/6	4/3/0
	30	(w/t/l)	0/0/7	0/0/7	0/0/7	0/0/7	0/0/7	3/0/4	0/0/7	0/0/7	4/0/3
	50	(w/t/l)	0/0/7	0/0/7	0/0/7	0/0/7	1/0/6	2/0/4	0/0/7	0/0/7	4/0/3

Table 4
The comparison of obtained solutions for hybrid functions.

F	D	Index	GWO	WOA	SSA	HHO	CoDE	EPSDE	QUATRE	MKE	MTDE
F11	10	Mean	2.2286E+01	1.0023E+02	8.0598E+01	5.6203E+01	4.6208E+00	2.6283E+00	2.3461E+00	8.8155E−01	**4.0190E−04**
		STD	1.2791E+01	5.8982E+01	5.9044E+01	4.5569E+01	9.6386E−01	9.4409E−01	1.9303E+00	1.1084E+00	5.1375E−04
		Min	7.5000E+00	2.4824E+01	2.6678E+01	1.0249E+01	3.2589E+00	6.1853E−01	9.9496E−01	5.4570E−12	1.4231E−06
	30	Mean	3.1822E+02	3.2385E+02	1.7555E+02	1.5694E+02	6.7047E+01	2.2479E+01	3.2970E+01	6.1087E+01	**1.2937E+01**
		STD	2.0078E+02	7.5186E+01	5.4067E+01	4.6287E+01	2.2958E+01	6.2666E+00	2.2884E+01	3.7682E+01	3.5522E+00
		Min	1.4792E+02	1.9604E+02	7.8608E+01	7.9862E+01	4.0090E+01	1.5351E+01	9.9496E+00	8.1316E+00	5.9698E+00
	50	Mean	1.2273E+03	4.9999E+02	2.4831E+02	2.9638E+02	1.0339E+02	7.7848E+01	1.3734E+02	1.1455E+02	**6.1268E+01**
		STD	9.4525E+02	1.0180E+02	6.5684E+01	6.3428E+01	2.1474E+01	4.4850E+01	5.8905E+01	3.4246E+01	1.2389E+01
		Min	3.6300E+02	3.3911E+02	1.4571E+02	1.6948E+02	4.5064E+01	3.6505E+01	3.5818E+01	4.9053E+01	3.8164E+01
F12	10	Mean	9.1104E+05	2.5032E+06	9.0603E+05	8.8754E+05	2.3852E+02	1.1449E+02	1.6622E+02	2.7519E+02	**2.8924E+00**
		STD	1.8098E+06	2.6979E+06	1.1882E+06	9.6292E+05	6.4220E+01	1.2048E+02	1.2728E+02	1.1553E+02	1.3644E+00
		Min	1.0946E+04	5.3531E+03	1.2457E+04	6.7988E+03	1.0043E+02	2.2737E−13	2.0814E−01	5.3463E+01	3.6904E−01
	30	Mean	3.1509E+07	4.2886E+07	2.6321E+06	7.5710E+06	2.7697E+03	4.4399E+04	1.8646E+04	1.8583E+04	**5.9356E+02**
		STD	2.8507E+07	2.9422E+07	2.3769E+06	3.9982E+06	1.1869E+03	4.0282E+04	1.5414E+04	1.2733E+04	2.5290E+02
		Min	1.6488E+06	1.1685E+07	4.0208E+05	1.9013E+06	1.7488E+03	5.2265E+03	9.2117E+02	4.5961E+03	1.6901E+02
	50	Mean	4.7401E+08	1.8504E+08	1.7676E+07	4.9198E+07	1.0415E+05	2.4173E+05	7.7601E+04	9.7152E+04	**3.0091E+04**
		STD	6.1872E+08	1.0098E+08	1.0333E+07	2.5930E+07	6.9660E+04	1.3197E+05	5.9357E+04	4.1716E+04	1.3645E+04
		Min	2.9100E+07	3.8786E+07	4.3290E+06	1.5231E+07	1.6963E+04	1.1197E+04	5.7463E+03	3.2218E+04	1.0972E+04
F13	10	Mean	9.7509E+03	1.4700E+04	1.3848E+04	1.6561E+04	1.2532E+01	6.4981E+00	6.1001E+00	6.7761E+00	**3.0457E−01**
		STD	6.5840E+03	1.1075E+04	1.1700E+04	9.2700E+03	3.0473E+00	1.8990E+00	3.3844E+00	4.1799E+00	3.7698E−01
		Min	1.4303E+03	1.1762E+03	6.4133E+02	1.5565E+03	7.4973E+00	4.6265E+00	9.9496E−01	3.1167E−03	1.5469E−02
	30	Mean	8.6219E+04	1.1017E+05	1.0485E+05	1.9681E+05	1.0195E+02	2.3759E+02	1.8898E+02	1.1364E+02	**2.9087E+01**
		STD	6.9184E+04	4.5900E+04	8.4103E+04	9.9206E+04	8.3170E+00	4.0022E+02	4.0319E+02	4.3133E+01	5.4712E+00
		Min	2.7580E+04	3.6418E+04	1.5522E+04	4.5553E+04	9.0540E+01	3.7236E+01	3.3207E+01	7.5494E+01	1.8188E+01
	50	Mean	8.3780E+07	1.5515E+05	1.2547E+05	1.1106E+06	2.5492E+02	5.5350E+03	1.5696E+03	6.1778E+03	**1.9730E+02**
		STD	1.0390E+08	1.1176E+05	6.2941E+04	3.1181E+05	3.8334E+01	8.7027E+03	5.5483E+03	6.6758E+03	6.3532E+01
		Min	5.0800E+04	4.1772E+04	4.5951E+04	5.7959E+05	2.0247E+02	1.0836E+02	1.2169E+02	2.8883E+02	9.8173E+01
F14	10	Mean	9.8408E+02	1.8732E+02	8.4108E+01	1.0387E+02	5.3971E+00	2.0295E+01	1.4041E+01	7.8341E−01	**1.9941E−01**
		STD	1.6069E+03	1.7130E+02	2.6246E+01	2.3936E+01	1.5316E+00	1.5818E−01	1.0726E+01	5.8186E−01	4.0811E−01
		Min	5.7897E+01	5.6458E+01	3.6087E+01	7.0586E+01	3.0280E+00	2.0005E+01	9.9496E−01	0.0000E+00	0.0000E+00
	30	Mean	1.1374E+05	9.6327E+05	6.2212E+03	3.1545E+04	5.1127E+01	4.4341E+01	8.2465E+01	5.0237E+01	**1.6754E+01**
		STD	2.8220E+05	1.5151E+06	4.5326E+03	4.1122E+04	5.7722E+00	9.7770E+00	4.4033E+01	1.0859E+01	8.9157E+00
		Min	1.6069E+03	3.3875E+04	5.6532E+02	4.2285E+02	3.9767E+01	3.4851E+01	3.9008E+01	2.8072E+01	4.9851E+00
	50	Mean	4.7498E+05	5.5527E+05	4.7767E+04	2.3352E+05	9.4249E+01	2.6301E+02	2.0952E+02	1.3242E+02	**7.1277E+01**
		STD	7.0945E+05	4.0930E+05	3.7779E+04	1.5090E+05	1.0959E+01	1.3701E+02	9.8001E+01	4.4682E+01	9.3593E+00
		Min	3.4100E+04	1.2059E+05	4.9087E+03	2.4971E+04	7.1616E+01	6.8590E+01	9.3972E+01	9.9541E+01	5.1942E+01
F15	10	Mean	1.4771E+03	2.7740E+03	5.2667E+02	5.7004E+02	1.8836E+00	1.3564E+00	1.6739E+00	3.1119E−01	**1.9996E−02**
		STD	1.7152E+03	3.4991E+03	6.0351E+02	5.5685E+02	5.1299E−01	1.3767E+00	1.0889E+00	5.3556E−01	1.1371E−02
		Min	2.5815E+01	1.2467E+02	3.9946E+01	7.8986E+01	1.0165E+00	3.1892E−03	1.7410E−03	7.3667E−04	3.0812E−03
	30	Mean	2.3397E+05	1.0406E+05	5.9988E+04	5.8160E+04	3.5426E+01	6.0091E+01	5.6813E+01	4.0869E+01	**1.2434E+01**
		STD	5.8191E+05	1.0358E+05	8.0953E+04	4.6172E+04	4.4603E+00	5.7412E+01	4.6813E+01	3.0879E+01	5.4498E+00
		Min	1.5830E+04	1.0181E+04	6.8435E+03	1.3508E+04	2.7413E+01	1.5994E+01	1.1946E+01	4.6319E+00	5.5109E+00
	50	Mean	3.5922E+06	8.3442E+04	4.8619E+04	1.8194E+05	**8.3172E+01**	2.0757E+02	2.1749E+02	1.1934E+02	9.1311E+01
		STD	7.4502E+06	6.3157E+04	3.0835E+04	6.5873E+04	1.3706E+01	1.8749E+02	9.4210E+01	2.4546E+01	2.2915E+01
		Min	2.3300E+04	2.0641E+04	1.2666E+04	1.0151E+05	5.5631E+01	5.1770E+01	5.0929E+01	8.7883E+01	4.2744E+01
F16	10	Mean	4.4638E+01	2.0043E+02	9.9448E+01	2.5059E+02	4.2855E+00	1.0306E+01	3.9560E+01	7.4335E+00	**2.7440E−01**
		STD	3.9803E+01	1.4516E+02	8.7031E+01	1.3010E+02	1.2383E+00	7.7254E+00	5.9041E+01	2.6364E+01	1.1200E−01
		Min	8.3635E+00	9.6021E+00	3.8970E+00	6.1199E+00	2.8863E+00	1.5635E+00	2.2795E−01	2.2881E−02	2.2961E−02
	30	Mean	6.9067E+02	1.8136E+03	8.7164E+02	1.3819E+03	7.2150E+02	7.0673E+02	8.4277E+02	3.8381E+02	**2.0457E+02**
		STD	2.1689E+02	5.0735E+02	2.7111E+02	3.4404E+02	1.2724E+02	1.5639E+02	2.5047E+02	2.2007E+02	1.1813E+02
		Min	2.5808E+02	1.0138E+03	1.9384E+02	7.1979E+02	4.9697E+02	3.7320E+02	3.0125E+01	7.7623E+00	1.1077E+01
	50	Mean	1.2685E+03	3.3390E+03	1.5126E+03	2.4596E+03	1.5548E+03	1.4253E+03	1.5064E+03	1.2441E+03	**7.4311E+02**
		STD	3.9197E+02	7.4499E+02	3.5162E+02	5.7540E+02	3.2176E+02	1.8526E+02	4.2760E+02	4.6771E+02	2.0099E+02
		Min	5.7900E+02	1.7146E+03	8.6269E+02	1.3655E+03	6.7374E+02	1.1281E+03	5.0615E+02	1.4007E+02	2.3543E+02
F17	10	Mean	5.1096E+01	9.0137E+01	5.9014E+01	7.9461E+01	1.4827E+01	2.1911E+01	1.7558E+01	8.6667E+00	**7.3652E−01**
		STD	2.6040E+01	4.3818E+01	2.2013E+01	6.6393E+01	3.2089E+00	2.7686E+00	1.3576E+01	1.7383E+01	4.1294E−01
		Min	2.0043E+01	3.1971E+01	2.6469E+01	3.5937E+01	6.9950E+00	1.2870E+01	1.3071E+00	0.0000E+00	9.9607E−03
	30	Mean	2.3266E+02	7.4875E+02	3.9262E+02	8.0600E+02	1.5240E+02	2.1984E+02	1.3412E+02	1.1473E+02	**5.3641E+01**
		STD	1.2344E+02	2.5902E+02	1.6774E+02	2.7542E+02	2.8635E+01	7.5929E+01	1.0065E+02	5.8993E+01	2.6076E+01
		Min	7.0739E+01	3.8325E+02	1.4222E+02	3.3256E+02	1.0675E+02	1.1726E+02	3.1261E+01	4.0230E+01	3.0695E+01
	50	Mean	9.9761E+02	2.3952E+03	1.4246E+03	2.0210E+03	1.0168E+03	9.5300E+02	9.7754E+02	8.3614E+02	**3.7358E+02**
		STD	2.4768E+02	3.4685E+02	2.3929E+02	4.3500E+02	1.4791E+02	1.3575E+02	2.3868E+02	2.2811E+02	1.0405E+02
		Min	4.4100E+02	1.6325E+03	1.0522E+03	1.3411E+03	6.7455E+02	7.1292E+02	2.3338E+02	3.0011E+02	8.6556E+01
F18	10	Mean	2.6070E+04	1.8298E+04	1.5054E+04	1.8956E+04	6.0608E+00	2.0439E+01	1.5120E+01	2.8010E+00	**7.9505E−03**
		STD	1.5208E+04	1.3315E+04	8.9580E+03	9.8520E+03	1.5870E+00	4.5799E−01	9.8001E+00	6.0468E+00	4.6019E−03

(continued on next page)

Table 4 (*continued*).

F	D	Index	GWO	WOA	SSA	HHO	CoDE	EPSDE	QUATRE	MKE	MTDE
		Min	9.5703E+02	2.2270E+02	1.9310E+03	3.5947E+03	2.0523E+00	2.0036E+01	2.1722E−01	4.4416E−03	1.8007E−03
	30	Mean	6.5272E+05	1.8884E+06	1.5947E+05	6.5640E+05	4.1172E+01	1.7520E+02	6.2332E+01	6.6587E+01	**2.8953E+01**
		STD	6.8499E+05	1.9841E+06	1.0493E+05	5.1296E+05	3.8535E+00	1.7046E+02	2.6438E+01	3.3198E+01	6.2953E+00
		Min	3.0608E+04	1.1048E+05	2.9133E+04	1.1073E+05	3.3150E+01	4.0848E+01	3.2048E+01	3.4866E+01	8.0825E+00
	50	Mean	2.6971E+06	5.2563E+06	4.7965E+05	1.8904E+06	**7.2239E+01**	3.5380E+03	5.5461E+03	3.3118E+03	9.6823E+01
		STD	3.9582E+06	5.0881E+06	2.5740E+05	1.5562E+06	1.9483E+01	2.5516E+03	4.4016E+03	2.2058E+03	3.6176E+01
		Min	3.9700E+05	6.0885E+05	1.6639E+05	3.8571E+05	4.0194E+01	3.7584E+02	3.4865E+02	8.6435E+02	5.5361E+01
F19	10	Mean	2.2156E+03	1.4674E+04	3.4042E+02	9.6920E+03	1.0553E+00	8.1841E−01	1.2000E+00	5.1071E−01	**1.7601E−02**
		STD	4.4939E+03	1.4334E+04	6.4615E+02	8.9253E+03	2.1126E−01	3.8142E−01	9.0128E−01	7.5911E−01	6.0010E−03
		Min	1.6449E+01	1.9280E+02	1.6449E+01	1.7838E+02	6.2384E−01	3.1823E−01	1.9432E−02	0.0000E+00	1.9360E−06
	30	Mean	5.1055E+05	2.6078E+06	3.8250E+05	9.9600E+04	2.6432E+01	2.2180E+01	2.6434E+01	1.4378E+01	**7.9013E+00**
		STD	4.7986E+05	2.8071E+06	2.1110E+05	7.5723E+04	3.2017E+00	5.9345E+00	1.8717E+01	5.2092E+00	1.5821E+00
		Min	9.6942E+03	3.3925E+04	1.3825E+04	3.6362E+03	2.0847E+01	1.3116E+01	9.8454E+00	7.7354E+00	5.2083E+00
	50	Mean	3.1386E+06	2.2840E+06	7.5369E+05	4.9492E+05	4.8586E+01	6.0124E+01	1.0377E+02	5.1441E+01	**4.1374E+01**
		STD	7.3260E+06	1.9068E+06	2.9493E+05	3.3903E+05	7.5682E+00	3.5633E+01	3.6805E+01	1.7772E+01	9.5732E+00
		Min	3.7900E+04	2.6724E+04	2.4449E+05	6.2907E+04	3.3916E+01	3.7594E+01	4.3070E+01	2.6249E+01	2.6849E+01
F20	10	Mean	5.8560E+01	1.1318E+02	1.0161E+02	1.3498E+02	3.8230E−01	1.6656E+01	7.9604E+00	6.1116E+00	**4.6826E−02**
		STD	5.0893E+01	5.1960E+01	7.6328E+01	6.1002E+01	3.9879E−01	4.4646E+00	1.0232E+01	1.0325E+01	1.1436E−01
		Min	2.3391E+00	4.7321E+01	2.4387E+01	4.4181E+01	1.5393E−02	6.8946E+00	3.1217E−01	0.0000E+00	0.0000E+00
	30	Mean	3.9264E+02	6.4772E+02	4.3423E+02	7.7234E+02	1.4081E+02	1.5686E+02	2.3982E+02	1.2325E+02	**5.4567E+01**
		STD	1.7362E+02	1.9879E+02	1.4266E+02	1.6226E+02	5.6671E+01	5.9538E+01	1.7314E+02	7.9963E+01	4.5344E+01
		Min	1.5773E+02	2.3146E+02	2.1620E+02	5.3482E+02	5.9990E+01	8.4106E+01	2.2238E+01	9.5890E+00	2.4777E+00
	50	Mean	8.0327E+02	1.6588E+03	9.6940E+02	1.4193E+03	6.3017E+02	7.2468E+02	9.1000E+02	6.8888E+02	**3.1922E+02**
		STD	2.4705E+02	2.3094E+02	3.1879E+02	3.6242E+02	1.8103E+02	1.4633E+02	2.5951E+02	2.5025E+02	1.4370E+02
		Min	4.0400E+02	1.3367E+03	3.6531E+02	7.1068E+02	1.8662E+02	3.1617E+02	1.0756E+02	8.1763E+01	1.1166E+02
Ranking	**10**	(w/t/l)	0/0/2	0/0/2	0/0/2	0/0/2	0/0/2	0/0/2	0/0/2	0/0/2	10/0/0
	30	(w/t/l)	0/0/10	0/0/10	0/0/10	0/0/10	0/0/10	0/0/10	0/0/10	0/0/10	10/0/0
	50	(w/t/l)	0/0/10	0/0/10	0/0/10	0/0/10	2/0/8	0/0/10	0/0/10	0/0/10	8/0/2

Table 5
The comparison of obtained solutions for composition functions.

F	D	Index	GWO	WOA	SSA	HHO	CoDE	EPSDE	QUATRE	MKE	MTDE
F21	10	Mean	2.0734E+02	2.0710E+02	1.4711E+02	1.7617E+02	1.1775E+02	1.8218E+02	1.6255E+02	1.6831E+02	**1.0000E+02**
		STD	2.2944E+01	5.9152E+01	5.7895E+01	6.9319E+01	4.3285E+01	4.6525E+01	5.8143E+01	5.7253E+01	0.0000E+00
		Min	1.1155E+02	1.0732E+02	1.0000E+02	1.0012E+02	1.0000E+02	1.0000E+02	1.0000E+02	1.0000E+02	1.0000E+02
	30	Mean	2.7619E+02	4.4932E+02	3.1317E+02	4.3642E+02	3.3282E+02	2.5983E+02	2.7077E+02	3.1219E+02	**2.2134E+02**
		STD	3.2223E+01	6.4638E+01	3.4683E+01	4.5564E+01	9.2272E+00	8.1311E+00	1.7519E+01	1.1020E+01	4.7374E+00
		Min	2.3466E+02	3.3804E+02	2.5940E+02	3.6371E+02	3.1319E+02	2.3575E+02	2.2613E+02	2.8723E+02	2.1292E+02
	50	Mean	3.7305E+02	7.5783E+02	4.2785E+02	6.7749E+02	4.8818E+02	3.8355E+02	3.5397E+02	4.5089E+02	**2.3993E+02**
		STD	2.6742E+01	9.7087E+01	4.0610E+01	8.4167E+01	2.3002E+01	1.3380E+01	3.7182E+01	1.5257E+01	9.1910E+00
		Min	3.2400E+02	6.0442E+02	3.6544E+02	5.3547E+02	4.4029E+02	3.4714E+02	2.9465E+02	4.1509E+02	2.2340E+02
F22	10	Mean	1.0426E+02	1.1416E+02	9.2645E+01	1.1175E+02	4.8366E+01	1.0011E+02	9.7546E+01	9.5905E+01	**1.2313E+01**
		STD	2.8976E+00	1.7460E+01	2.4838E+01	1.7186E+01	5.0332E+01	1.0774E−01	1.7838E+01	2.2583E+01	3.0351E+01
		Min	1.0107E+02	5.0401E+01	1.8453E+01	4.2082E+01	3.4503E−02	1.0000E+02	1.1563E+01	4.5475E−13	0.0000E+00
	30	Mean	2.1293E+03	3.7227E+03	2.1525E+03	3.8035E+03	1.0000E+02	2.2921E+03	8.7699E+02	1.5373E+03	**1.0000E+02**
		STD	1.5304E+03	2.2536E+03	1.9527E+03	1.9258E+03	1.1585E−03	2.0079E+03	1.4028E+03	2.2605E+03	2.3218E−13
		Min	1.5816E+02	1.1209E+02	1.0000E+02	1.1633E+02	1.0000E+02	3.9279E+02	1.0000E+02	1.0000E+02	1.0000E+02
	50	Mean	6.1380E+03	9.6954E+03	7.1884E+03	8.4836E+03	9.2072E+03	9.9494E+03	7.1606E+03	9.3656E+03	**5.0889E+03**
		STD	6.9282E+02	1.2223E+03	1.2934E+03	8.2988E+02	3.9375E+03	4.7529E+02	1.1670E+03	6.8115E+02	5.4047E+02
		Min	5.0500E+03	8.0735E+03	5.6768E+03	7.0655E+03	1.0001E+02	8.4045E+03	6.0982E+03	7.2883E+03	4.1194E+03
F23	10	Mean	3.1352E+02	3.4121E+02	3.2192E+02	3.5153E+02	3.1393E+02	3.1380E+02	3.0909E+02	3.0749E+02	**3.0366E+02**
		STD	7.2011E+00	1.3550E+01	9.0964E+00	1.8118E+01	1.8298E+00	2.5971E+00	4.5598E+00	2.8501E+00	1.3398E+00
		Min	3.0573E+02	3.1678E+02	3.1060E+02	3.1652E+02	3.0964E+02	3.0801E+02	3.0000E+02	3.0000E+02	3.0003E+02
	30	Mean	4.3866E+02	7.0604E+02	4.5278E+02	7.6406E+02	4.7846E+02	4.1019E+02	4.0929E+02	4.5517E+02	**3.7496E+02**
		STD	3.9338E+01	7.4467E+01	3.1411E+01	1.1921E+02	8.1707E+00	8.9888E+00	1.9613E+01	8.6869E+00	3.4801E+00
		Min	3.8362E+02	5.5597E+02	4.0739E+02	5.7944E+02	4.6465E+02	3.9654E+02	3.7074E+02	4.3820E+02	3.6541E+02
	50	Mean	6.2448E+02	1.2849E+03	6.6191E+02	1.2722E+03	7.3105E+02	5.8430E+02	5.8391E+02	6.7669E+02	**4.8196E+02**
		STD	3.6534E+01	1.5258E+02	4.9193E+01	1.4780E+02	1.4732E+01	2.2328E+01	3.7432E+01	2.1305E+01	1.2304E+01
		Min	5.7900E+02	1.0436E+03	5.9744E+02	9.5191E+02	7.0180E+02	5.5512E+02	5.0574E+02	6.4462E+02	4.5221E+02
F24	10	Mean	3.4260E+02	3.1616E+02	3.4740E+02	3.8463E+02	**1.9772E+02**	3.3682E+02	3.1662E+02	2.9222E+02	2.5683E+02
		STD	9.4487E+00	1.2169E+02	9.3846E+00	9.5167E+01	1.2280E+02	1.7198E+00	7.4229E+01	9.8640E+01	1.1841E+02
		Min	3.3119E+02	2.5671E+01	3.3099E+02	1.0114E+02	1.0000E+02	3.3323E+02	1.0000E+02	1.0000E+02	0.0000E+00
	30	Mean	4.9881E+02	7.6078E+02	5.1192E+02	9.5031E+02	5.5194E+02	4.7990E+02	4.8054E+02	5.2898E+02	**4.5508E+02**
		STD	4.5834E+01	8.5578E+01	2.1263E+01	1.1307E+02	1.4432E+01	1.3235E+01	1.6560E+01	1.1511E+01	4.7161E+00
		Min	4.4900E+02	6.0605E+02	4.8005E+02	7.2846E+02	5.1455E+02	4.5333E+02	4.4540E+02	5.0767E+02	4.4560E+02

(*continued on next page*)

Table 5 (*continued*).

F	D	Index	GWO	WOA	SSA	HHO	CoDE	EPSDE	QUATRE	MKE	MTDE
	50	Mean	6.9264E+02	1.2749E+03	7.0994E+02	1.6702E+03	6.6404E+02	6.7883E+02	6.5350E+02	7.4593E+02	**5.5521E+02**
		STD	6.5735E+01	1.3829E+02	4.0963E+01	2.5899E+02	9.2112E+01	1.4604E+01	3.8139E+01	2.0008E+01	8.9082E+00
		Min	6.2600E+02	1.0325E+03	6.2571E+02	1.2342E+03	5.4512E+02	6.4311E+02	5.9388E+02	7.0753E+02	5.4092E+02
F25	10	Mean	4.3590E+02	4.4375E+02	4.2257E+02	4.2292E+02	3.9813E+02	4.1885E+02	4.1647E+02	4.2107E+02	**3.9786E+02**
		STD	1.5001E+01	3.1465E+01	2.5855E+01	8.2088E+01	2.4484E−01	2.3775E+01	2.3116E+01	2.3672E+01	1.3570E−01
		Min	4.0009E+02	4.0001E+02	3.9774E+02	1.0983E+02	3.9774E+02	3.9774E+02	3.9774E+02	3.9774E+02	3.9774E+02
	30	Mean	4.4925E+02	4.4409E+02	4.0344E+02	4.0747E+02	3.8690E+02	**3.7840E+02**	3.8689E+02	3.8631E+02	3.8680E+02
		STD	3.1003E+01	2.9979E+01	2.3391E+01	2.0898E+01	3.8462E−02	1.0058E−01	1.0351E+00	1.2550E+00	7.8082E−01
		Min	3.9168E+02	3.9405E+02	3.8342E+02	3.8454E+02	3.8681E+02	3.7825E+02	3.8344E+02	3.8340E+02	3.8351E+02
	50	Mean	8.9866E+02	6.2678E+02	5.3084E+02	6.2048E+02	4.8909E+02	**4.4578E+02**	5.0924E+02	5.1266E+02	5.0017E+02
		STD	1.8009E+02	4.9376E+01	3.6815E+01	3.2269E+01	2.5263E+01	1.8465E+01	3.8172E+01	4.2407E+01	2.8412E+01
		Min	7.0300E+02	5.2639E+02	4.6045E+02	5.4905E+02	4.7892E+02	4.3115E+02	4.7580E+02	4.7737E+02	4.6082E+02
F26	10	Mean	4.0041E+02	6.0399E+02	3.0707E+02	8.0837E+02	3.0000E+02	3.1775E+02	3.0707E+02	**2.9500E+02**	3.0000E+02
		STD	2.9728E+02	3.5628E+02	1.7273E+01	4.4302E+02	2.4138E−05	5.4626E+01	1.7273E+01	2.2361E+01	0.0000E+00
		Min	2.2913E+02	2.0470E+02	3.0000E+02	3.0113E+02	3.0000E+02	3.0000E+02	2.0000E+02	2.0000E+02	3.0000E+02
	30	Mean	1.8317E+03	4.7373E+03	1.6828E+03	4.2821E+03	2.0861E+03	1.2901E+03	1.6900E+03	1.9153E+03	**1.1629E+03**
		STD	2.1218E+02	1.0434E+03	1.0363E+03	1.1589E+03	5.8629E+02	8.1369E+01	1.9157E+02	3.9972E+02	6.2452E+01
		Min	1.4424E+03	1.6310E+03	2.0000E+02	3.1362E+02	3.0667E+02	1.1702E+03	3.0000E+02	3.0000E+02	1.0539E+03
	50	Mean	3.2515E+03	1.0668E+04	1.9575E+03	6.1672E+03	3.9185E+03	2.7638E+03	2.6624E+03	3.4849E+03	**1.6379E+03**
		STD	4.7688E+02	1.6055E+03	1.9141E+03	3.1281E+03	1.2547E+02	4.4986E+02	4.3749E+02	1.7496E+02	1.2539E+02
		Min	2.2800E+03	7.9367E+03	3.0000E+02	3.3469E+02	3.6289E+03	1.9944E+03	2.0620E+03	3.1628E+03	1.3760E+03
F27	10	Mean	3.9377E+02	4.1310E+02	3.9148E+02	4.3425E+02	**3.8800E+02**	3.9868E+02	3.9740E+02	3.9181E+02	**3.8800E+02**
		STD	2.8834E+00	2.4058E+01	2.4533E+00	3.8449E+01	7.8984E−01	4.4155E+01	1.8611E+01	2.9871E+00	9.2093E−01
		Min	3.8964E+02	3.9465E+02	3.8931E+02	3.9846E+02	3.8695E+02	3.7082E+02	3.8901E+02	3.8689E+02	3.8689E+02
	30	Mean	5.4047E+02	6.4268E+02	5.2891E+02	6.1610E+02	5.0292E+02	**5.0001E+02**	5.0175E+02	5.0073E+02	5.0500E+02
		STD	2.1716E+01	7.2930E+01	1.7648E+01	5.0238E+01	1.0320E+01	1.0004E−04	1.0940E+01	1.1233E+01	6.0335E+00
		Min	5.0955E+02	5.4000E+02	4.9914E+02	5.3384E+02	4.7708E+02	5.0001E+02	4.8926E+02	4.7728E+02	4.8714E+02
	50	Mean	7.8897E+02	1.3721E+03	6.8093E+02	1.2121E+03	5.1296E+02	**5.0001E+02**	5.7880E+02	5.4000E+02	6.0242E+02
		STD	7.0112E+01	2.0665E+02	6.0535E+01	1.9971E+02	1.0427E+01	7.5281E−05	7.5822E+01	2.5388E+01	2.9033E+01
		Min	6.3200E+02	9.7955E+02	5.8444E+02	9.0094E+02	5.0040E+02	5.0001E+02	5.1752E+02	5.0423E+02	5.5280E+02
F28	10	Mean	5.6252E+02	5.4985E+02	4.1080E+02	4.2578E+02	**3.0000E+02**	4.7444E+02	4.0212E+02	3.4537E+02	**3.0000E+02**
		STD	9.0391E+01	1.7947E+02	1.0134E+02	1.1142E+02	3.9984E−04	3.8978E+00	1.4299E+02	1.1093E+02	0.0000E+00
		Min	3.6900E+02	3.0039E+02	3.0000E+02	3.0049E+02	3.0000E+02	4.7250E+02	3.0000E+02	3.0000E+02	3.0000E+02
	30	Mean	5.5470E+02	4.9063E+02	4.1814E+02	4.3720E+02	3.2351E+02	4.9963E+02	3.3619E+02	3.5564E+02	**3.0917E+02**
		STD	6.0837E+01	2.2024E+01	4.3535E+01	1.8136E+01	3.7048E+01	1.2179E+00	5.7332E+01	5.8460E+01	2.8344E+01
		Min	4.7288E+02	4.3838E+02	3.0000E+02	4.0356E+02	3.0134E+02	4.9506E+02	3.0000E+02	3.0000E+02	3.0000E+02
	50	Mean	1.1505E+03	6.3395E+02	4.9340E+02	5.3928E+02	**4.5885E+02**	5.0001E+02	4.8550E+02	4.8014E+02	4.7396E+02
		STD	3.4736E+02	5.2765E+01	2.2151E+01	3.3751E+01	1.3338E−03	1.6168E−02	2.3284E+01	2.4333E+01	2.1704E+01
		Min	6.7200E+02	5.6989E+02	4.5895E+02	4.7500E+02	4.5885E+02	4.9994E+02	4.5885E+02	4.5885E+02	4.5885E+02
F29	10	Mean	2.6670E+02	3.6804E+02	2.6850E+02	3.9833E+02	2.6436E+02	2.5310E+02	2.5216E+02	2.6209E+02	**2.3855E+02**
		STD	3.8435E+01	6.8970E+01	2.5486E+01	7.7687E+01	6.8734E+00	7.4809E+00	1.5739E+01	2.8961E+01	2.7011E+00
		Min	2.3346E+02	2.7237E+02	2.3595E+02	2.6282E+02	2.5328E+02	2.3967E+02	2.3163E+02	2.4385E+02	2.3412E+02
	30	Mean	7.9601E+02	1.9230E+03	9.7302E+02	1.3444E+03	6.9362E+02	5.6080E+02	6.2085E+02	5.3200E+02	**4.5202E+02**
		STD	1.6964E+02	3.5050E+02	1.6512E+02	2.8205E+02	4.3894E+01	1.0191E+02	1.8161E+02	7.5620E+01	2.6399E+01
		Min	4.8916E+02	1.0564E+03	7.8147E+02	7.3729E+02	5.7927E+02	3.5949E+02	4.2708E+02	4.1605E+02	3.8430E+02
	50	Mean	1.3255E+03	3.7718E+03	1.7909E+03	2.4183E+03	1.0833E+03	1.1130E+03	9.1367E+02	7.3547E+02	**4.7271E+02**
		STD	2.7462E+02	8.3557E+02	2.3790E+02	4.0706E+02	1.0630E+02	1.9076E+02	2.7894E+02	2.1046E+02	8.7990E+01
		Min	7.6300E+02	2.0630E+03	1.3031E+03	1.7206E+03	8.3082E+02	7.8403E+02	5.3017E+02	4.4473E+02	3.4410E+02
F30	10	Mean	6.7964E+05	2.8394E+05	1.8434E+05	5.8792E+05	5.2309E+02	**2.0802E+02**	2.8874E+05	3.2786E+05	3.9555E+02
		STD	8.1531E+05	4.2661E+05	3.4875E+05	7.8780E+05	3.4446E+01	8.1781E+00	4.6595E+05	4.1024E+05	2.4986E−01
		Min	6.6548E+02	3.2415E+03	5.0901E+03	2.8771E+03	4.6177E+02	2.0147E+02	3.9465E+02	6.7373E+02	3.9507E+02
	30	Mean	4.9767E+06	9.9049E+06	1.9479E+06	8.7251E+05	2.1931E+03	**2.1883E+02**	2.2163E+03	3.1430E+03	2.0915E+03
		STD	4.9154E+06	6.4555E+06	1.4352E+06	3.7817E+05	4.8220E+01	3.6568E+00	2.2863E+02	2.5523E+03	4.3592E+01
		Min	2.0469E+05	1.2098E+06	1.7755E+05	3.3649E+05	2.1237E+03	2.1347E+02	1.9632E+03	2.1457E+03	2.0216E+03
	50	Mean	7.5521E+07	8.8362E+07	2.7491E+07	1.3045E+07	6.0434E+05	**8.7379E+02**	6.8421E+05	7.4471E+05	6.3127E+05
		STD	2.1339E+07	2.9338E+07	5.7171E+06	3.1455E+06	1.1167E+04	1.5203E+03	1.0961E+05	1.0695E+05	2.9077E+04
		Min	4.1800E+07	4.5857E+07	2.0504E+07	8.0018E+06	5.8950E+05	2.8124E+02	5.7941E+05	6.0722E+05	5.5824E+05
Ranking	**10**	(w/t/l)	0/0/2	0/0/2	0/0/2	0/0/2	1/2/7	1/0/9	0/0/10	1/0/9	5/2/3
	30	(w/t/l)	0/0/10	0/0/10	0/0/10	0/0/10	0/1/9	3/0/7	0/0/10	0/0/10	6/1/3
	50	(w/t/l)	0/0/10	0/0/10	0/0/10	0/0/10	1/0/9	3/0/7	0/0/10	0/0/10	6/0/4

4.5. Statistical analysis

Although the results obtained from the quantitative analysis show that the MTDE algorithm outperforms the comparative algorithms, the rank of the algorithms in these experiments is not determined. Therefore, to prove the superiority of MTDE is statistically significant, two non-parametric tests are conducted: Friedman and Wilcoxon signed-rank tests.

4.5.1. Non-parametric Friedman test

In the first statistical test, the Friedman test (F_f) [75] is used for ranking MTDE and other algorithms based on their obtained fitness using Eq. (15) where k is the number of algorithms, R_j

Table 6
Effectiveness of the MTDE and other state-of-the-art competitor algorithms.

	GWO (w/t/l)	WOA (w/t/l)	SSA (w/t/l)	HHO (w/t/l)	CoDE (w/t/l)	EPSDE (w/t/l)	QUATRE (w/t/l)	MKE (w/t/l)	MTDE (w/t/l)
D=10	0/0/29	0/0/29	0/0/29	0/0/29	2/1/26	2/1/26	0/4/25	0/4/25	20/7/2
D=30	0/0/29	0/0/29	0/0/29	0/0/29	0/1/28	6/1/22	0/1/28	0/1/28	21/2/6
D=50	0/0/29	0/0/29	0/0/29	0/0/29	4/0/25	6/0/23	0/0/29	0/0/29	19/0/10
Total	0/0/87	0/0/87	0/0/87	0/0/87	6/2/79	14/2/71	0/5/82	0/5/82	60/9/18
OE	0%	0%	0%	0%	9.1%	18.3%	5.7%	5.7%	**79.3%**

Table 7
Overall rank by Friedman test in dimension D = 10, 30 and 50.

Alg.	D	F_1	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}	F_{16}	F_{17}
	10	7.45	8.35	8.15	4.70	6.05	5.20	4.50	6.70	4.90	6.50	7.00	7.15	7.25	7.70	6.20	6.70
GWO	30	9.00	7.90	8.00	4.10	6.00	4.55	3.65	6.00	2.40	7.90	8.00	6.85	7.70	7.20	4.10	5.10
	50	9.00	7.40	8.65	3.45	6.00	5.30	3.70	6.20	2.15	8.90	8.55	8.00	7.80	7.75	3.75	4.60
	10	7.85	8.60	7.55	8.55	8.40	8.50	8.25	8.65	8.40	8.15	8.20	7.60	8.30	8.40	7.80	8.15
WOA	30	7.10	8.90	8.35	8.70	8.80	8.30	8.75	8.60	7.95	8.60	8.55	7.50	8.85	8.15	8.75	8.25
	50	7.10	7.75	8.00	8.35	8.80	8.25	8.55	8.65	6.50	8.00	8.30	6.90	8.60	7.30	8.75	8.70
	10	6.00	4.00	6.55	6.50	7.15	6.40	6.90	6.05	6.40	7.95	7.35	7.50	7.00	6.90	6.95	7.30
SSA	30	5.90	2.70	6.05	5.75	7.05	4.90	6.00	7.15	4.45	6.95	6.10	7.15	6.35	7.10	6.15	6.70
	50	5.55	**1.00**	5.55	6.20	7.00	5.10	6.15	6.85	3.60	6.15	6.05	6.70	6.15	6.55	4.90	6.95
	10	8.70	7.05	7.75	8.30	8.40	8.25	7.95	8.35	7.40	7.40	7.45	7.75	7.45	7.00	8.05	7.65
HHO	30	7.90	5.90	7.10	8.20	8.15	8.70	7.40	8.25	6.40	6.45	7.35	8.50	7.10	7.55	7.75	8.45
	50	7.90	5.25	7.15	8.15	8.20	8.75	7.95	8.30	4.60	6.75	7.10	8.40	7.45	8.40	7.95	8.15
	10	5.00	6.00	4.15	5.65	5.00	5.95	5.80	5.25	5.60	4.75	3.95	4.70	3.35	4.30	3.65	3.20
CoDE	30	5.10	4.75	5.40	6.50	5.00	6.70	7.00	4.50	8.25	4.50	2.05	3.95	3.50	3.50	4.60	3.80
	50	5.45	3.10	3.40	6.65	4.80	6.45	6.65	1.90	8.75	3.40	3.25	2.70	2.05	**1.65**	5.60	4.45
	10	**2.00**	**2.00**	**1.65**	2.10	2.42	2.35	2.15	**2.50**	3.60	3.50	2.50	2.95	4.50	3.30	4.50	4.55
EPSDE	30	3.40	3.20	**1.00**	2.50	**1.07**	2.30	2.25	**1.55**	5.20	2.55	4.50	3.50	2.65	3.65	4.55	4.95
	50	**1.00**	7.60	**1.45**	3.30	**1.00**	2.65	2.95	**1.73**	7.00	2.25	4.50	3.95	4.35	3.50	4.15	3.85
	10	**2.00**	**2.00**	2.70	3.40	2.15	3.25	4.00	**2.50**	3.35	3.15	3.15	2.95	3.98	3.95	3.75	3.90
Quatre	30	2.25	3.75	3.58	2.90	2.98	3.30	3.40	2.75	3.05	2.95	3.75	2.55	4.40	3.65	5.45	3.20
	50	2.70	4.00	3.15	2.50	3.70	2.50	2.65	4.50	4.20	4.15	2.50	**1.90**	4.10	4.35	4.85	4.00
	10	4.00	5.00	4.85	4.35	3.38	4.05	4.30	**2.50**	4.25	2.05	4.20	3.30	1.93	2.05	2.60	2.00
MKE	30	3.23	6.90	2.92	5.35	2.08	5.25	5.55	3.25	6.15	3.80	3.70	3.90	3.45	3.10	2.25	3.10
	50	4.00	6.90	3.75	5.40	2.00	5.00	5.40	2.52	7.05	3.70	3.30	4.45	3.35	3.30	3.80	3.20
	10	**2.00**	**2.00**	**1.65**	**1.45**	**2.05**	**1.05**	**1.15**	**2.50**	**1.10**	**1.55**	**1.20**	**1.10**	**1.25**	**1.40**	**1.50**	**1.55**
MTDE	30	**1.13**	**1.00**	2.60	**1.00**	3.88	**1.00**	**1.00**	2.95	**1.15**	**1.30**	**1.00**	**1.10**	**1.00**	**1.10**	**1.40**	**1.45**
	50	2.30	2.00	3.90	**1.00**	3.50	**1.00**	**1.00**	4.35	**1.15**	**1.70**	**1.45**	2.00	**1.15**	2.20	**1.25**	**1.10**

Alg.		F_{18}	F_{19}	F_{20}	F_{21}	F_{22}	F_{23}	F_{24}	F_{25}	F_{26}	F_{27}	F_{28}	F_{29}	F_{30}	Avg. rank	Overall rank
	10	8.00	6.65	6.40	6.30	6.50	4.60	5.10	6.45	6.50	6.20	7.95	4.70	7.15	6.45	7
GWO	30	7.30	7.35	6.05	3.65	5.95	4.40	3.60	8.10	4.65	6.75	8.65	5.75	7.90	6.16	7
	50	7.70	7.50	4.45	3.10	2.55	4.05	4.25	9.00	4.75	6.80	9.00	5.55	8.45	6.15	7
	10	7.25	8.30	7.80	7.80	8.20	8.15	6.80	7.80	7.85	7.95	7.15	8.25	6.90	7.98	9
WOA	30	8.40	8.75	7.80	8.30	7.50	8.45	8.05	8.25	8.40	8.55	7.40	8.80	8.55	8.32	9
	50	8.70	8.35	8.75	8.75	6.80	8.40	8.15	7.45	8.95	8.75	7.95	8.95	8.55	8.17	9
	10	7.00	6.65	7.55	4.85	5.60	6.55	6.45	4.80	5.20	5.05	5.55	4.95	6.35	6.33	6
SSA	30	6.40	7.60	6.85	5.30	5.45	5.15	4.80	5.70	4.55	6.15	5.10	6.80	7.10	5.98	6
	50	6.10	7.45	5.65	4.90	3.85	5.05	5.05	5.00	3.40	6.00	5.25	7.00	7.00	5.59	6
	10	7.75	8.40	8.05	6.75	8.15	8.60	8.05	6.65	8.60	8.55	6.05	8.50	7.35	7.81	8
HHO	30	7.90	6.30	8.60	8.60	7.70	8.55	8.95	6.75	8.00	8.20	5.50	8.00	6.45	7.61	8
	50	7.50	6.70	7.70	8.25	5.30	8.60	8.85	7.25	6.80	8.25	6.50	7.95	6.00	7.45	8
	10	3.25	3.90	2.35	3.85	3.65	5.00	3.45	3.75	5.65	**2.20**	3.55	5.30	3.50	4.33	5
CoDE	30	2.45	4.70	3.15	6.70	3.80	6.60	6.75	3.75	6.10	3.15	3.20	4.95	3.65	4.76	5
	50	**1.40**	2.75	3.55	6.70	7.65	6.90	4.05	3.55	6.80	2.20	**2.55**	4.35	2.50	4.32	5
	10	4.60	3.35	4.50	4.90	3.10	5.15	4.10	4.55	2.88	2.75	6.65	3.85	**1.00**	3.38	2
EPSDE	30	4.50	3.65	3.45	2.55	6.40	2.95	3.00	**1.00**	2.35	**2.55**	7.80	2.75	**1.00**	3.20	2
	50	4.00	2.85	4.40	3.70	7.35	2.75	3.90	**1.35**	3.50	**1.00**	4.35	4.40	**1.00**	3.44	2
	10	3.95	3.90	3.73	4.15	4.75	3.20	4.67	4.67	3.05	5.17	3.63	3.60	4.55	3.56	3
Quatre	30	3.45	3.40	4.60	3.35	2.33	2.75	2.95	4.05	3.85	2.95	2.25	3.80	3.15	3.34	3
	50	4.20	4.75	5.10	2.80	3.60	2.75	3.40	3.90	3.45	4.05	3.25	3.35	3.55	3.58	3
	10	2.20	2.30	3.40	5.15	3.95	2.65	4.53	4.80	3.10	4.30	3.00	4.40	6.05	3.61	4
MKE	30	3.55	2.20	2.95	5.55	4.33	5.15	5.75	2.85	5.40	2.85	3.15	2.60	4.80	3.97	4
	50	3.80	2.90	3.95	5.80	6.55	5.50	6.35	4.00	5.60	3.15	2.90	2.35	4.55	4.29	4
	10	**1.00**	**1.55**	**1.23**	**1.25**	**1.10**	**1.10**	**1.85**	**1.52**	**2.17**	2.83	**1.48**	**1.45**	2.15	**1.56**	**1**
MTDE	30	**1.05**	**1.05**	**1.55**	**1.00**	**1.55**	**1.00**	**1.15**	4.55	**1.70**	3.85	**1.95**	**1.55**	2.40	**1.67**	**1**
	50	1.60	**1.75**	**1.45**	**1.00**	**1.35**	**1.00**	**1.00**	3.50	**1.75**	4.80	3.25	**1.10**	3.40	**2.00**	**1**

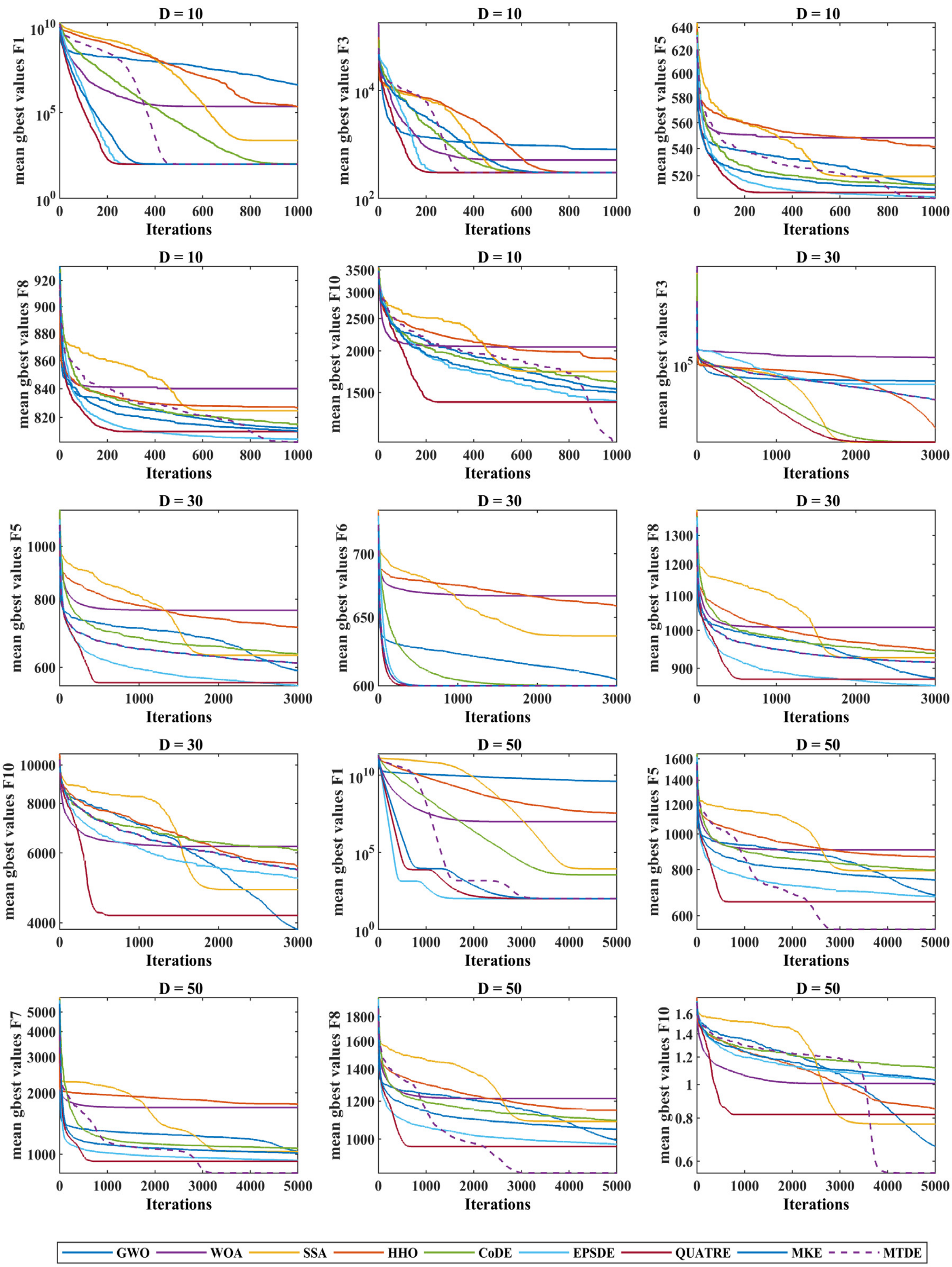

Fig. 9. Convergence evaluation for unimodal and multimodal benchmark functions with different dimensions.

is the mean rank of the j-th algorithm, n is the number of case tests. The test is performed by assuming χ^2 distribution with k-1 degrees of freedom. It first finds the rank of algorithms individually and then calculates the average rank to get the final rank of each algorithm for the considered problem.

$$F_f = \frac{12n}{k(k+1)}\left[\sum_j R_j^2 - \frac{k(k+1)^2}{4}\right] \tag{15}$$

Based on the results of the test shown in Table 7, there is a significant difference between MTDE and other algorithms

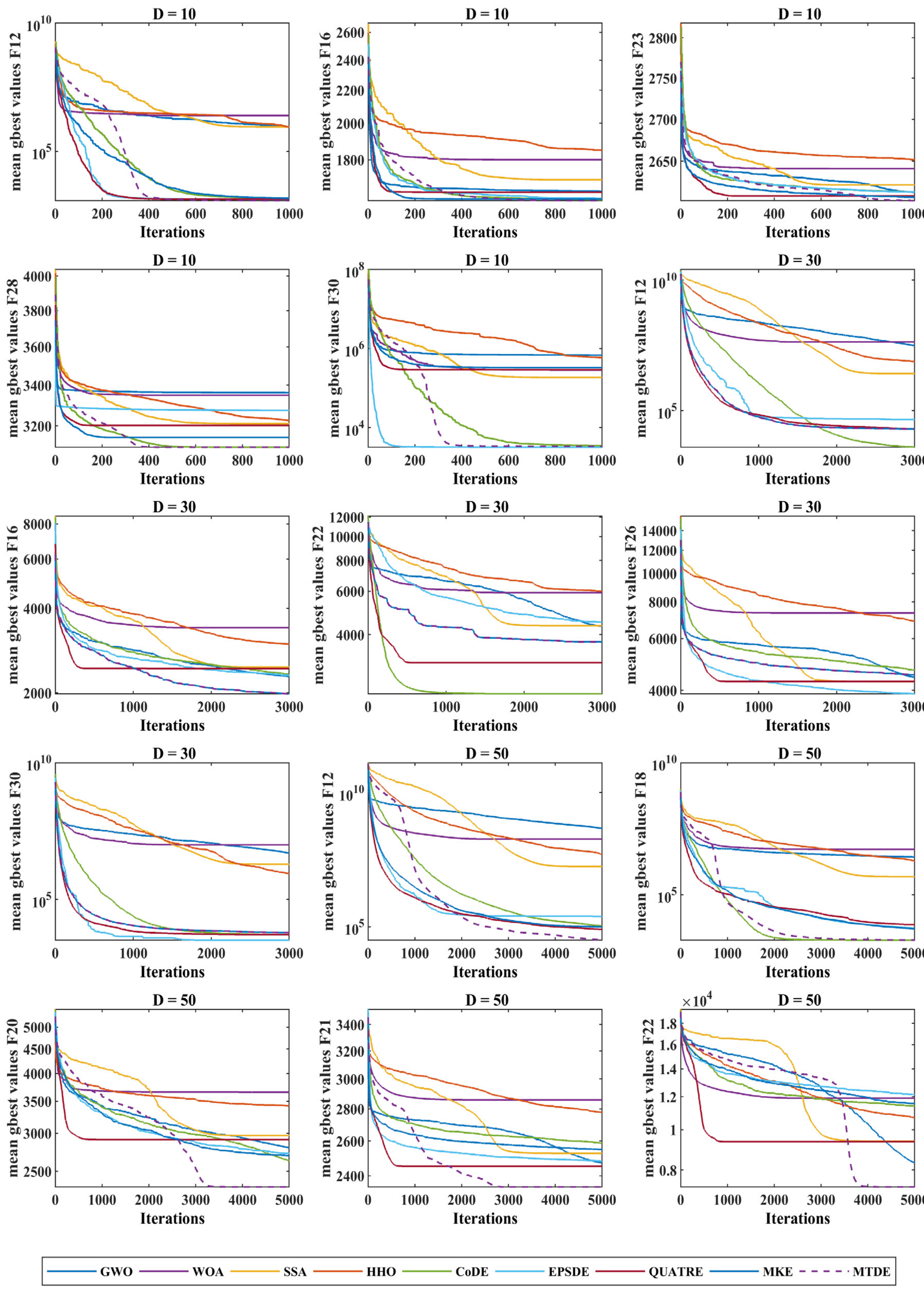

Fig. 10. Convergence evaluation for hybrid and composition functions with different dimensions.

for different functions. The overall rank shows that our MTDE algorithm is superior to other algorithms for all dimensions 10, 30, and 50.

4.5.2. Wilcoxon signed-rank test

In order to verify the significant difference between MTDE and other comparative algorithms, the Wilcoxon signed-rank test is employed [75]. Tables 8–10 presents the results of this test at

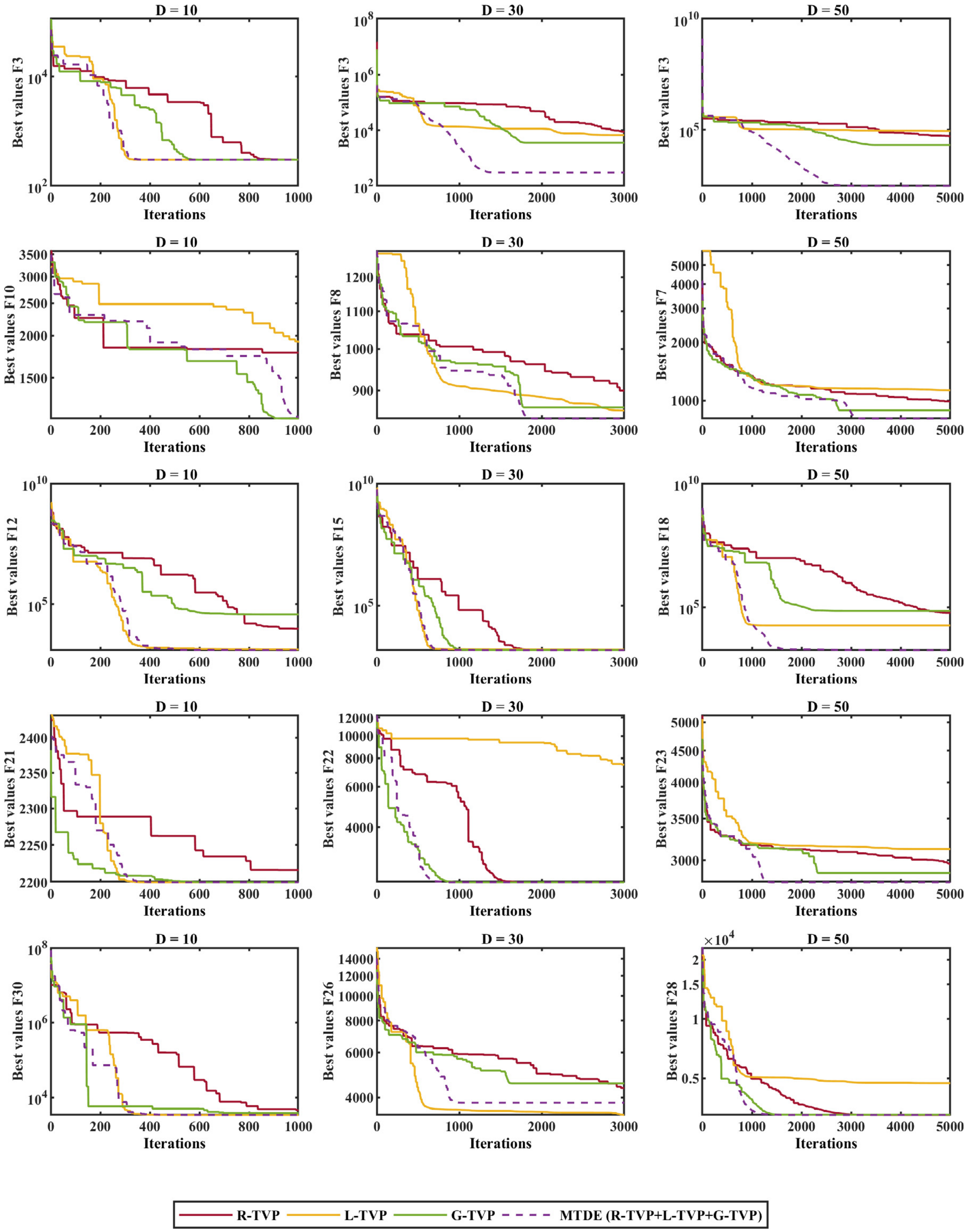

Fig. 11. The impact analysis and comparison of using individual TVPs and MTDE.

significance level $\alpha = 0.05$, where R^+ and R^- values represent the sum of ranks for the functions where MTDE performs better or worse than its competitor. The *p*-value refers to the significant difference between each pair of algorithms (MTDE and one other algorithm). The considerable difference exists only if *p*-value < α. The results indicate that MTDE is able to significantly outperform the state-of-the-art algorithms for the functions at all dimensions.

5. Solving engineering design problems using MTDE

In this section, the proposed MTDE algorithm has been applied on four engineering design problems: pressure vessel, welded beam, tension/compression spring, and three-bar truss. These problems have several equality and inequality constraints; therefore, MTDE should be able to handle them during the optimization process. There are several strategies of constraint handling in the literature: rejecting, penalizing, repairing, decoding, and preserving strategies [1]. The death penalty function is one of

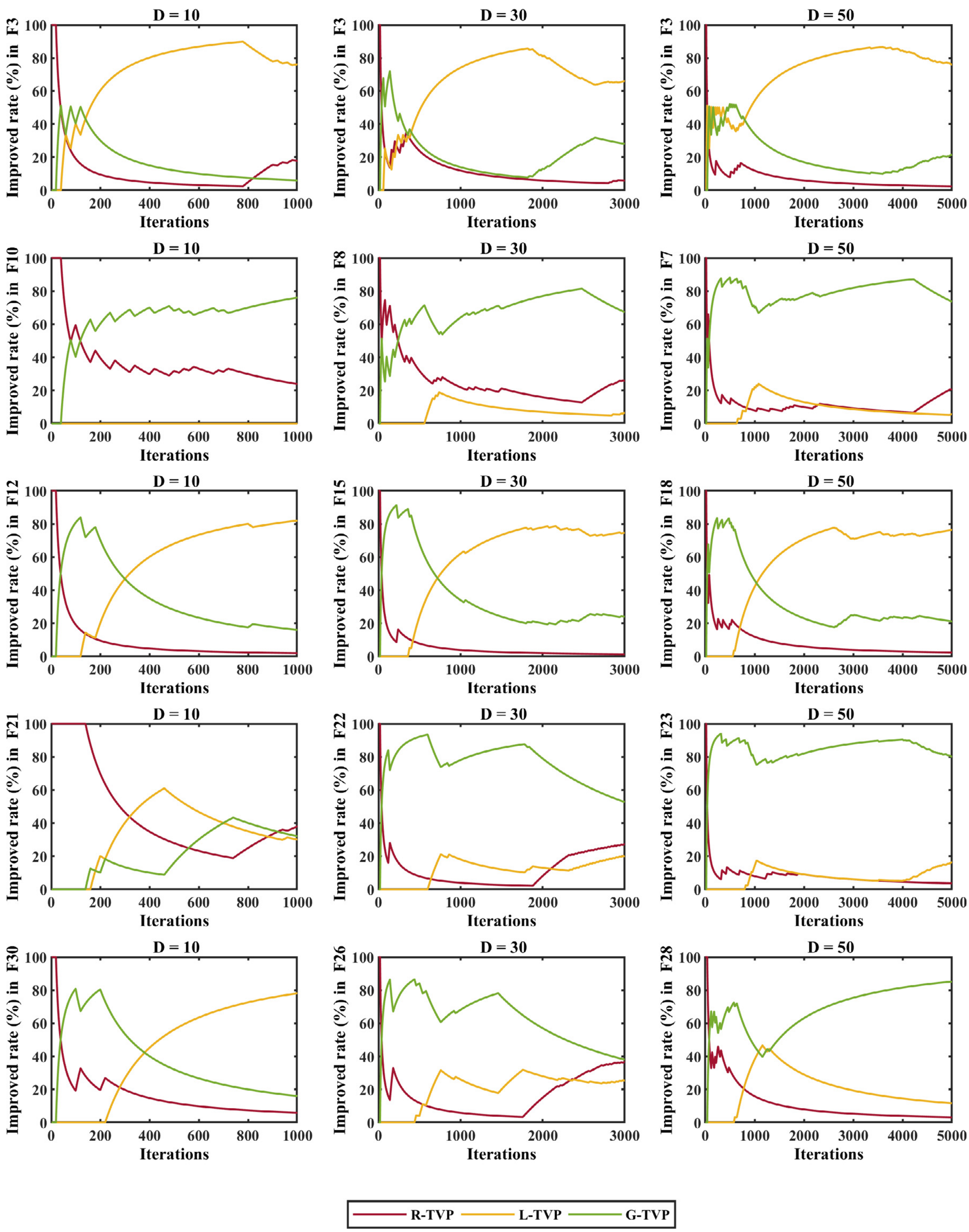

Fig. 12. The impact percentage of each TVP in the obtained solutions.

the simplest techniques among the different constraint handling strategies to handle the multi-constraint problems. This function assigns a large fitness value to those solutions that violate any of the constraints in order to discard the infeasible solutions.

In this experiment, the population size (N), the maximum number of iterations, and the number of runs were set to 20, $(D\times 10\text{\^{}}4)/N$, and 10, respectively. Tables 11–14 show that MTDE finds the best optimal values for variables of these engineering problems compared to other algorithms.

6. Discussion

In this section, the main reasons for the superiority of the MTDE algorithm over the comparative algorithms are discussed. Findings from the previous sections clearly demonstrate that using the introduced MTV search strategy can adapt the proposed MTDE algorithm and enhance its performance for different kinds of problems. From the results and curves shown in Table 2 and Fig. 9, the MTDE algorithm is competitive for unimodal problems, and it has a faster convergence than other algorithms. The main

Table 8
Results of Wilcoxon's test on D = 10.

MTDE vs.	R^+	R^-	*p*-value	$\alpha = 0.05$
GWO	435	0	2.56E−06	YES
WOA	435	0	2.563E−06	YES
SSA	435	0	2.563E−06	YES
HHO	435	0	2.563E−06	YES
CoDE	435	0	2.563E−06	YES
EPSDE	435	0	2.563E−06	YES
Quatre	435	0	2.563E−06	YES
MKE	435	0	2.563E−06	YES

Table 9
Results of Wilcoxon's test on D = 30.

MTDE vs.	R^+	R^-	*p*-value	$\alpha = 0.05$
GWO	435	0	2.56E−06	YES
WOA	435	0	2.56E−06	YES
SSA	435	0	2.56E−06	YES
HHO	435	0	2.56E−06	YES
CoDE	435	0	2.56E−06	YES
EPSDE	435	0	2.56E−06	YES
Quatre	435	0	2.56E−06	YES
MKE	435	0	2.56E−06	YES

Table 10
Results of Wilcoxon's test on D = 50.

MTDE vs.	R^+	R^-	*p*-value	$\alpha = 0.05$
GWO	435	0	2.56E−06	YES
WOA	435	0	2.563E−06	YES
SSA	434	1	2.849E−06	YES
HHO	435	0	2.563E−06	YES
CoDE	407	28	4.175E−05	YES
EPSDE	406	29	4.582E−05	YES
Quatre	435	0	2.563E−06	YES
MKE	435	0	2.563E−06	YES

Table 11
Results for the pressure vessel problem.

Alg.	Optimum values for variables				Optimum cost
	T_s	T_h	R	L	
GWO	0.778709	0.386125	40.341393	199.710960	5890.8880
WOA	0.795325	0.396759	40.755204	194.023927	5986.1040
SSA	0.804226	0.397532	41.669723	182.022972	5931.4000
HHO	0.893918	0.447576	46.301073	130.697786	6137.6940
CoDE	0.881838	0.449010	43.051343	168.197626	6532.4070
EPSDE	0.778169	0.384649	40.319619	200	5885.3330
QUATRE	0.831172	0.410849	43.065887	165.006244	5982.2440
MKE	0.778169	0.384649	40.319619	200	5885.3330
MTDE	0.778169	0.384649	40.319619	200	**5885.3328**

Table 12
Results for the welded beam problem.

Alg.	Optimum values for variables				Optimum cost
	h	l	t	b	
GWO	0.205409	3.478839	9.035941	0.205774	1.725700
WOA	0.189953	3.996270	8.710470	0.227041	1.871528
SSA	0.204620	3.494451	9.036845	0.205729	1.726388
HHO	0.182232	4.114156	8.991604	0.207795	1.779197
CoDE	0.200068	3.575454	9.226260	0.206701	1.770640
EPSDE	0.205730	3.470489	9.036624	0.205730	**1.724852**
QUATRE	0.205741	3.470296	9.036497	0.205742	1.724919
MKE	0.205729	3.470548	9.036550	0.205734	1.724876
MTDE	0.205730	3.470489	9.036624	0.205730	**1.724852**

reason is that unimodal problems are solved by using L-TVP more than other TVPs, which the local information of its individuals enhances the exploitation. Meanwhile, the distribution policy used in the MTDE algorithm provides more chances for the L-TVP.

Table 13
Results for the tension/compression spring design problem.

Alg.	Optimum values for variables			Optimum weight
	d	D	N	
GWO	0.051176	0.344493	12.058282	0.012684
WOA	0.054237	0.421179	8.314372	0.012779
SSA	0.050000	0.317217	14.055452	0.012733
HHO	0.051248	0.346188	11.934085	0.012669
CoDE	0.050264	0.319697	14.080980	0.012989
EPSDE	0.051689	0.356711	11.289385	**0.012665**
QUATRE	0.056957	0.497201	6.146576	0.013140
MKE	0.052466	0.375666	10.269711	0.012688
MTDE	0.051689	0.356718	11.288965	**0.012665**

Table 14
Results for the three-bar truss problem.

Alg.	Optimum values for variables		Optimum weight
	x1	x2	
GWO	0.787946	0.410323	2.63897000E+02
WOA	0.841256	0.276715	2.65614500E+02
SSA	0.788821	0.407837	2.63895900E+02
HHO	0.786062	0.415689	2.63900900E+02
CoDE	0.787621	0.411278	2.63900800E+02
EPSDE	0.788675	0.408248	**2.63895800E+02**
QUATRE	0.789514	0.405885	2.63896700E+02
MKE	0.791907	0.399584	2.63943500E+02
MTDE	0.788675	0.408248	**2.63895800E+02**

The results and curves in Table 3 and Fig. 9 prove the superiority of MTDE on the majority of multimodal functions. The main reason for this merit of the proposed algorithm in the exploration and fast convergence is using the G-TVP and its gbest-history. Using the gbests provides individuals to avoid local optima, which results in exploring the search space extensively. However, because of using the gbests in the G-TVP, in the distribution policy, we use the penalty rule for G-TVP in order to increase the diversity by other TVPs and avoid local optima.

The results are shown in Tables 4 and 5, and the convergence curves plotted in Fig. 10 indicate that the MTDE algorithm is competitive with other algorithms for solving hybrid and composition functions. Since these functions have been created using shifted and rotated unimodal and multimodal functions, the results demonstrate that MTDE can appropriately balance exploration and exploitation. Moreover, it maintains the diversity to handle difficulties in these kinds of complex functions. The main reason is that MTDE uses multi-trial vector producers (TVPs) designed for solving different problems. Based on the results obtained from TVPs in each *WinIter*, the more effective TVP is determined as Win-TVP for the next *WinIter* iterations. Therefore, the MTDE algorithm is not greedy to use only one TVP during all iterations, which guarantees to close the global optimum by proper TVPs during the course of iterations. Another reason that MTDE is competitive for hybrid and composition functions is using the combination of suitable TVPs which maintains the diversity, avoids premature convergence, and escapes from local optima.

In addition, using the unionPopulation in R-TVP and L-TVP and also considering the information of worst individuals in the R-TVP help to maintain the diversity and guide other individuals toward the best solution. Moreover, the ability to escape from local optima and approach the global optimum on the majority of functions are satisfactory since MTDE uses the winner-based distributing policy. Although we used a simple policy, it still has a significant impact on the performance of MTDE in dealing with different problems with various requirements.

As in all studies, there are some limitations of our study. The MTDE is proposed to solve single-objective and continuous problems, however, there exist many multi-objective and discrete problems. The winner-based distribution policy was designed based on three TVPs used in this study and it must be redesigned for new trial vectors. Also, when the dimension of a problem is extremely increased, the performance of the proposed MTDE may decrease. To avoid of this limitation, the size of the life-time archive and its policy must be determined for high dimension problems by some pre-experiments.

7. Conclusion and future works

In this article, we first introduced an MTV approach in which a combination of different search strategies can enhance the abilities of movement strategy, such as exploration and exploitation ability in different problems. In the MTV approach, the population is dynamically distributed between some subpopulations according to a winner-based distributing policy. Then, based on the introduced MTV approach, an effective differential evolution algorithm named MTDE is proposed. In MTDE, only three trial vector producers R-TVP, L-TVP, and G-TVP were designed to cope with different problems. Although, a simple distributing policy was used to get the TVP's experiences, it properly distributes the population between their subpopulations.

Meanwhile, MTDE uses a life-time archiving to keep the inferior solutions that have some information about the visited promising regions, which can enrich the quality of the population. Also, to avoid the local optima, MTDE provides gbest-history to keep the best obtained individuals, and the information about the high quality explored regions of the search space. Therefore, our proposed algorithm shares individuals' information during the optimization process using the distributing policy, the life-time archiving, and the gbest-history.

The experiment and statistical results and the above-mentioned discussions support the following conclusions:

- The introduced MTV approach provides using different trial vector producers concurrently by which the efficiency of movement strategy is enhanced for various problems.
- The MTV approach enables us to design and add another efficient TVPs when it is needed to cope with some challenges in solving new problems.
- In the MTV approach, the winner-based distributing step and considering *WinIter* iterations gives the opportunity of defining a distribution policy to respect the winner TVP in each *WinIter*.
- An effective DE algorithm based on the MTV approach was proposed using three introduced trial vector producers R-TVP, L-TVP, and G-TVP and a winner-based distributing policy.
- Using different trial vectors, life-time archiving, gbest-history, and union population provide information sharing between individuals in the MTDE algorithm.
- The experimental results and statistical tests prove that MTDE is more efficient than the comparative algorithms for different unimodal, multimodal, hybrid, and composition problems.
- The proposed MTDE is applicable for solving engineering design problems.

In this article, we used the MTV approach for DE algorithms, but it can be used for other categories of metaheuristics such as swarm intelligence algorithms. Also, although we introduced three trial vector producers R-TVP, L-TVP and G-TVP, and a simple distributing policy to develop the MTDE algorithm, introducing another combination of TVPs with effective distribution policy can be considered as future works. The MTDE algorithm is a single-objective algorithm designed for continuous problems; thus, the multi-objective and discrete versions can be implemented. Moreover, the applications of the proposed algorithm to solve real-world engineering problems are another possible future works.

CRediT authorship contribution statement

Mohammad H. Nadimi-Shahraki: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Supervision, Project administration. **Shokooh Taghian:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft, Writing - review & editing. **Seyedali Mirjalili:** Validation, Formal analysis, Writing - review & editing, Supervision. **Hossam Faris:** Validation, Formal analysis, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Four engineering design problems used in Section 4.5 are described as follows.

— Pressure vessel design problem

The main objective of pressure vessel design problem [76] shown in Fig. A.1 is to minimize the cost of material, forming, and welding of a vessel. Four variables of this problem are the thickness of shell (T_s) and head (T_h), inner radius (R), and cylindrical section length without considering the head (L). The mathematical formulation and four constraint functions of this problem are given in Eq. (A.1).

$$\begin{aligned}
\text{Consider} \quad & \vec{x} = [x_1 x_2 x_3 x_4] = [T_s T_h R L] \\
\text{Minimize} \quad & f(\vec{x}) = 0.6224 x_1 x_3 x_4 + 1.7781 x_2 x_3^2 \\
& + 3.1661 x_1^2 x_4 + 19.84 x_1^2 x_3 \\
\text{Subject to} \quad & g_1(\vec{x}) = -x_1 + 0.0193 x_3 \le 0, \\
& g_2(\vec{x}) = -x_2 + 0.00954 x_3 \le 0, \\
& g_3(\vec{x}) = -\pi x_3^2 x_4 - \frac{4}{3}\pi x_3^3 + 1.296.000 \le 0, \\
& g_4(\vec{x}) = x_4 - 240 \le 0 \\
\text{where} \quad & 0 \le x_i \le 100.i = 1.2 \\
& 10 \le x_i \le 200.i = 3.4
\end{aligned} \tag{A.1}$$

— Welded beam problem

The objective of this design problem is to obtain a minimum fabrication cost for designing a welded beam [77]. In Fig. A.2, there are four design variables to be optimized: the weld thickness (h), the attached part of the bar length (l), the bar height (t), and the bar thickness (b). Also, by applying the load on top of the bar, seven constraints should not be violated. These constraints are: shear stress (τ), bending stress in the beam (σ), end deflection of the beam (δ), and buckling load on the bar (P_b). The formulation of this problem is computed by Eq. (A.2).

$$\text{Consider} \quad \vec{x} = [x_1 x_2 x_3 x_4] = [hltb] \tag{A.2}$$

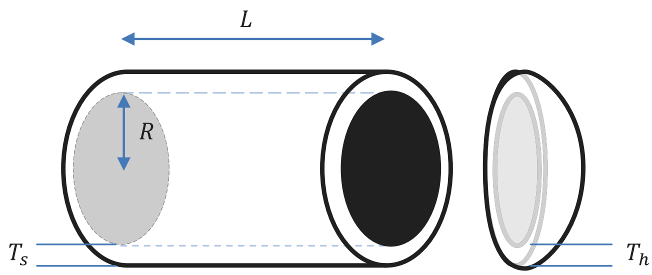

Fig. A.1. Design of pressure vessel problem.

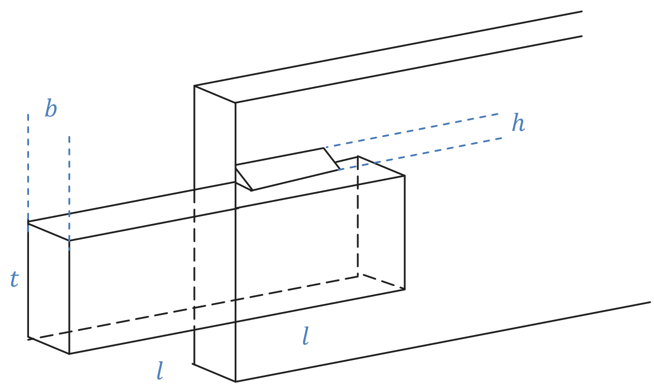

Fig. A.2. Design of welded beam problem.

$$
\begin{aligned}
\text{Minimize} \quad & f(\vec{x}) = 1.10471 x_1^2 x_2 + 0.04811 x_3 x_4 * (14.0 + x_2) \\
\text{Subject to} \quad & g_1(\vec{x}) = \tau(\vec{x}) - \tau_{max} \le 0, \\
& g_2(\vec{x}) = \sigma(\vec{x}) - \sigma_{max} \le 0, \\
& g_3(\vec{x}) = \delta(\vec{x}) - \delta_{max} \le 0, \\
& g_4(\vec{x}) = x_1 - x_4 \le 0 \\
& g_5(\vec{x}) = \mathrm{P} - P_c(\vec{x}) \le 0 \\
& g_6(\vec{x}) = 0.125 - x_1 \le 0 \\
& g_7(\vec{x}) = 1.10471 x_1^2 + 0.04811 x_3 x_4 \\
& \quad * (14.0 + x_2) - 0.5 \le 0 \\
\text{where} \quad & 0.1 \le x_i \le 2.i = 1.2 \\
& 0.1 \le x_i \le 10.i = 3.4
\end{aligned}
$$

– Tension/compression spring design problem

The main objective of this problem, as described in [78] is to minimize the tension/compression spring weight. As shown in Fig. A.3, this problem consists of three design variables: wire diameter (d), mean coil diameter (D), and the number of active coils (P). The mathematical representation of this problem shown in Eq. (A.3).

$$
\begin{aligned}
\text{Consider} \quad & \vec{x} = [x_1 x_2 x_3] = [dDN] \qquad \text{(A.3)} \\
\text{Minimize} \quad & f(\vec{x}) = (x_3 + 2) x_2 x_1^2 \\
\text{Subject to} \quad & g_1(\vec{x}) = 1 - \frac{x_2^3 x_3}{71785 x_1^4} \le 0, \\
& g_2(\vec{x}) = \frac{4x_2^2 - x_1 x_2}{12566(x_2 x_1^3 - x_1^4)} + \frac{1}{5108 x_1^2} \le 0, \\
& g_3(\vec{x}) = 1 - \frac{140.45 x_1}{x_2^2 x_3} \le 0, \\
& g_4(\vec{x}) = \frac{x_{1+} x_2}{1.5} - 1 \le 0 \\
\text{where} \quad & 0.05 \le x_1 \le 2.00,
\end{aligned}
$$

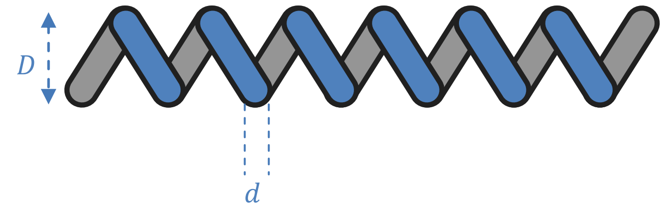

Fig. A.3. Design of tension/compression spring design problem.

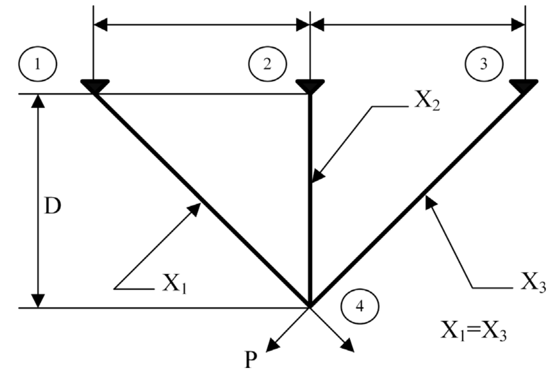

Fig. A.4. Design of three-bar truss problem.

$$
\begin{aligned}
& 0.25 \le x_2 \le 1.30, \\
& 2.00 \le x_3 \le 15.0
\end{aligned}
$$

– Three-bar truss problem

The objective of this problem is to design a truss with minimum weight, which does not violate three constraints [79]. With regard to Fig. A.4, two design variables X_1 and X_2 should be determined subject to stress, deflection, and buckling constraints. The formulation of this problem shown in Eq. (A.4).

$$
\begin{aligned}
\text{Consider} \quad & \vec{x} = [x_1 x_2] = [A_1 A_2] \qquad \text{(A.4)} \\
\text{Minimize} \quad & f(\vec{x}) = \left(2\sqrt{2} x_1 + x_2\right) * l \\
\text{Subject to} \quad & g_1(\vec{x}) = \frac{\sqrt{2} x_1 + x_2}{\sqrt{2} x_1^2 + 2 x_1 x_2} \mathrm{P} - \sigma \le 0, \\
& g_2(\vec{x}) = \frac{x_2}{\sqrt{2} x_1^2 + 2 x_1 x_2} \mathrm{P} - \sigma \le 0, \\
& g_3(\vec{x}) = \frac{1}{\sqrt{2} x_2 + x_1} \mathrm{P} - \sigma \le 0, \\
\text{where} \quad & 0 \le x_1.x_2 \le 1, \\
& l = 100 \text{ cm}.\ P = 2 \text{ kN/cm}^2, \sigma = 2 \text{ kN/cm}^2
\end{aligned}
$$

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